

Special Seminar on Multi-Modal AI

Jiayuan Mao

Department of CIS, GRASP

<https://cis7000-fall26.thelabone.org/>

Many slides adapted from <https://learning3d.github.io/>

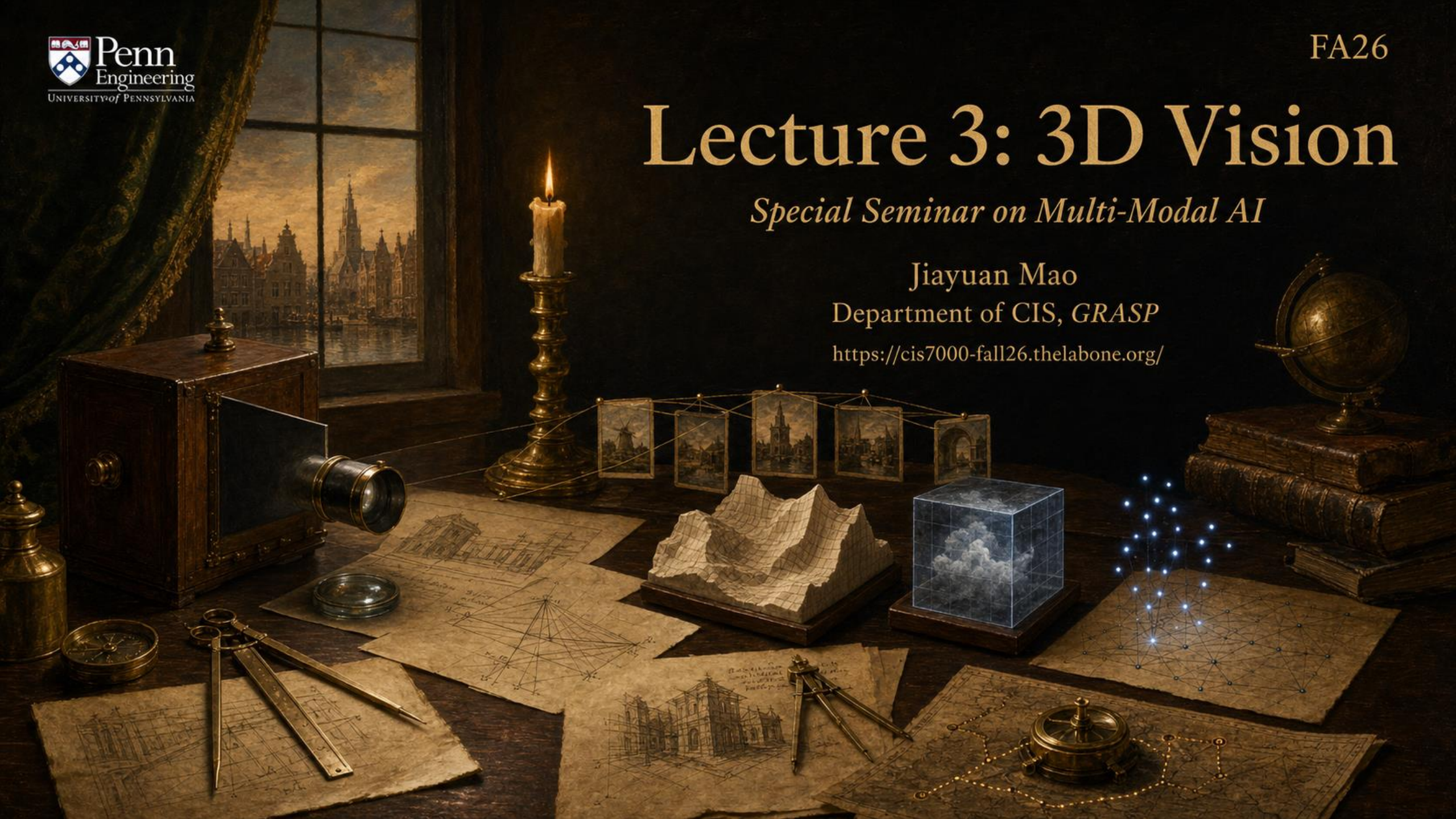
Lecture 3: 3D Vision

Special Seminar on Multi-Modal AI

Jiayuan Mao

Department of CIS, GRASP

<https://cis7000-fall26.thelabone.org/>



Enrollment

I believe I have approved all students who attended the first two lectures and confirmed that they are comfortable with the lecture format and the final project arrangement

Logistics

Check the course website for slides and recordings

<https://cis7000-fall26.thelabone.org>

Reminder of the final project

Final project

Projects may cover any topic related to multimodal AI, including topics connected to a student's own research field. Teams may include up to two students, and individual projects are welcome.

Master's theses and capstone projects are allowed with instructor approval. The project may also be a subset of a paper for publication, as long as it is connected to the course.

- **Sun Sep 20:** Initial proposal due.
- **Sun Oct 11:** Final proposal due.
- **Sun Nov 1:** Mid-project check-in through office hours.
- **Wed Dec 2 and Mon Dec 7:** Final project presentations.

Initial Project Proposal

Start Assignment

Due Sep 20 by 11:59pm **Points** 5 **Submitting** a file upload **File Types** doc, docx, and pdf

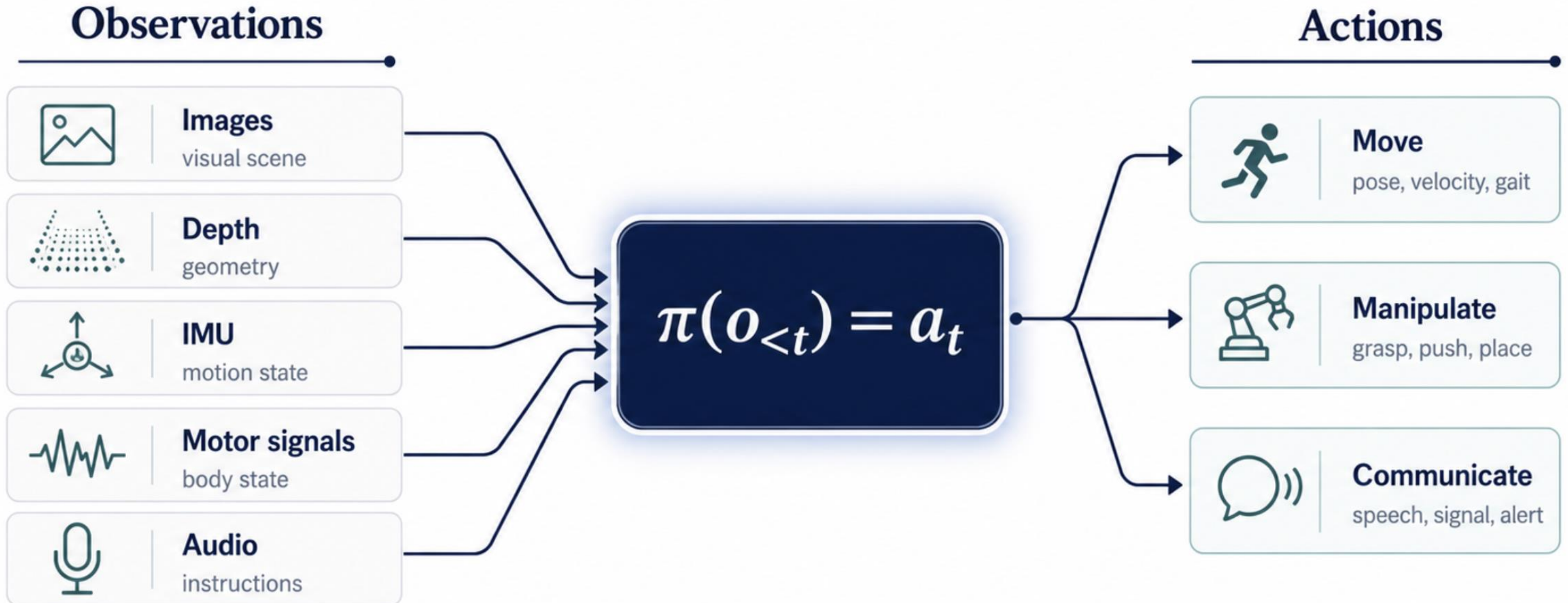
No additional details were added for this assignment.

Goal of This Course

Use Machine Learning to Build Intelligent Agents

Abstract View of An Agent: Multimodal AI

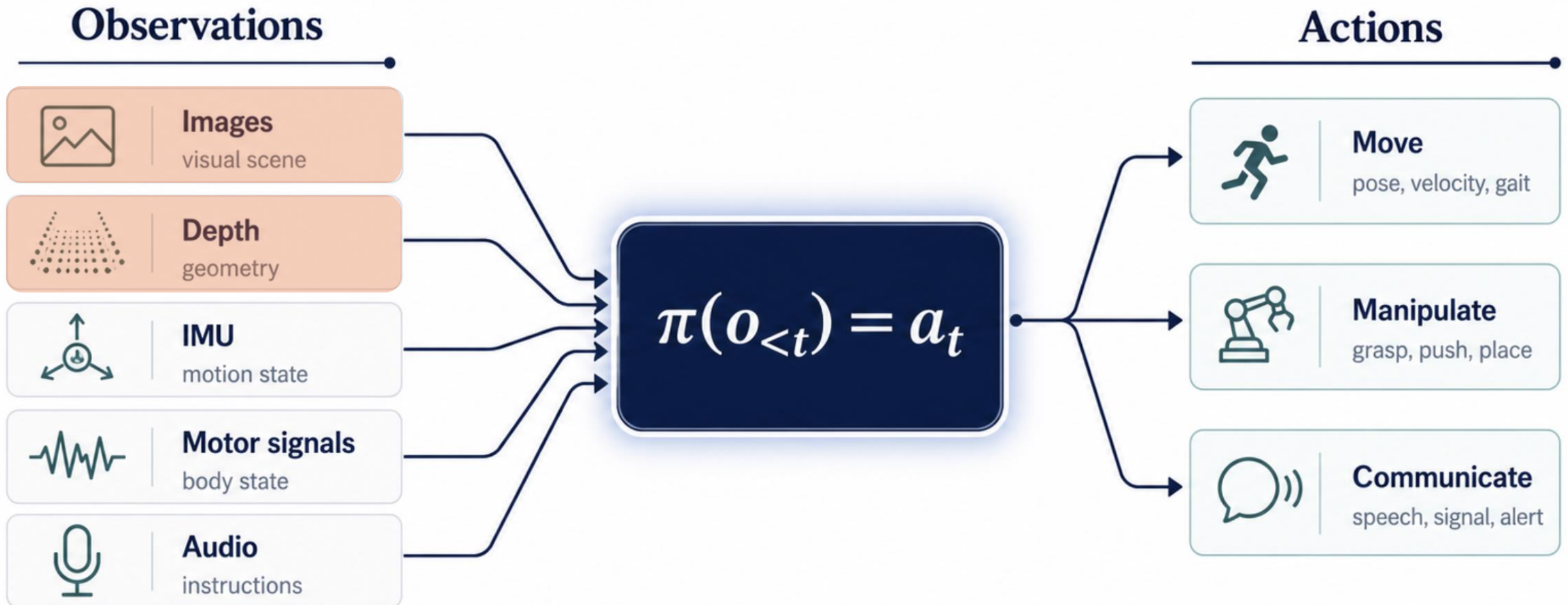
π : observations $\rightarrow$ action



Notation: t – time. $o_{<t}$ - all observations before t . a_t - action at t .

Abstract View of An Agent: Multimodal AI

π : observations $\rightarrow$ action



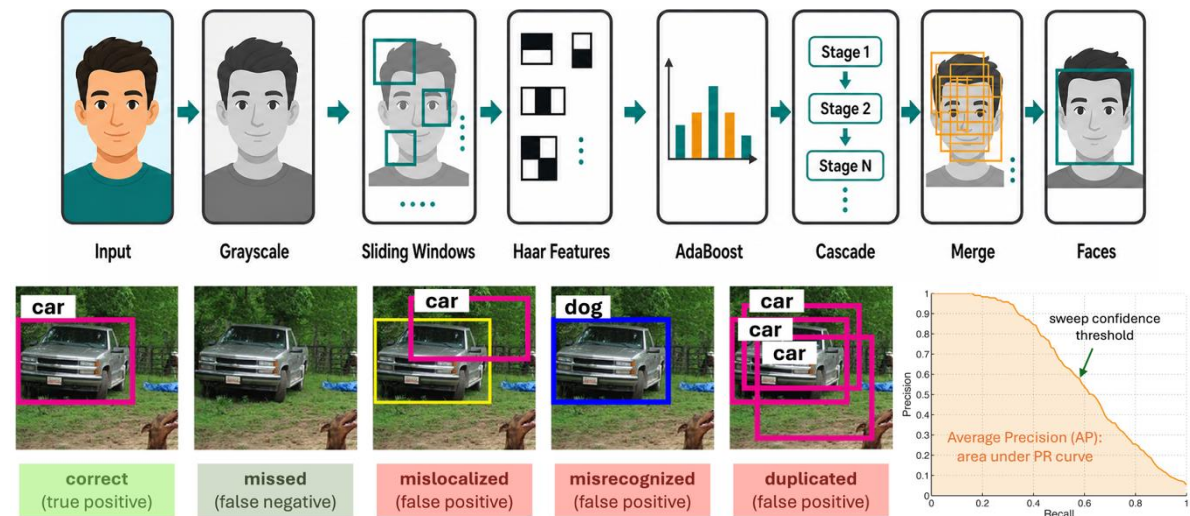
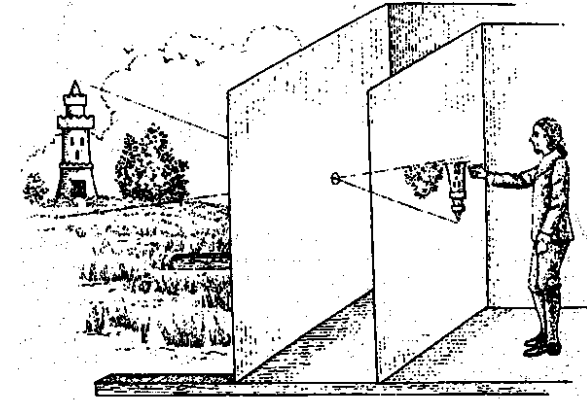
Notation: t – time. $o_{<t}$ - all observations before t . a_t - action at t .

Goal of This Lecture

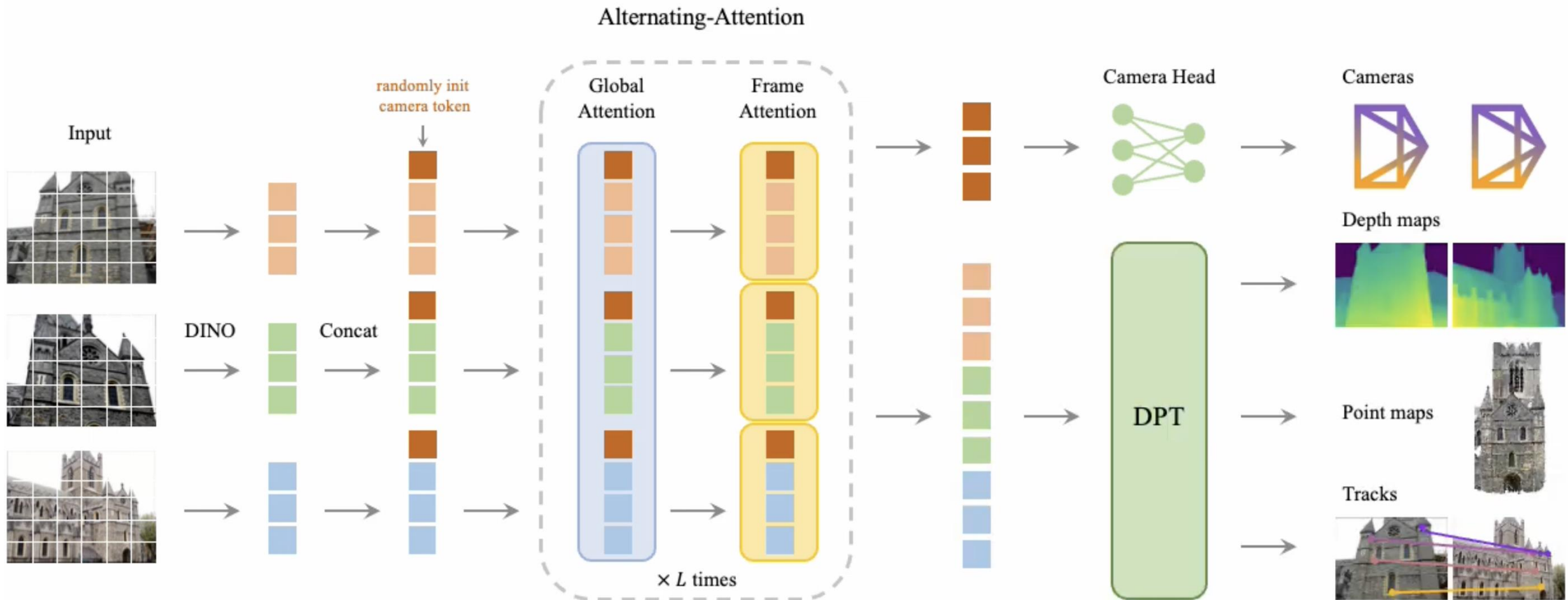
Use Machine Learning to Sense and Understand The 3D World

Connections with the Previous Lecture

- Understand Pinhole camera
 - Use the camera model as a natural supervision
- Understand principles of image recognition
 - Pattern matching, sliding window
 - Multi-stage prediction, result merging
 - Evaluation quirks

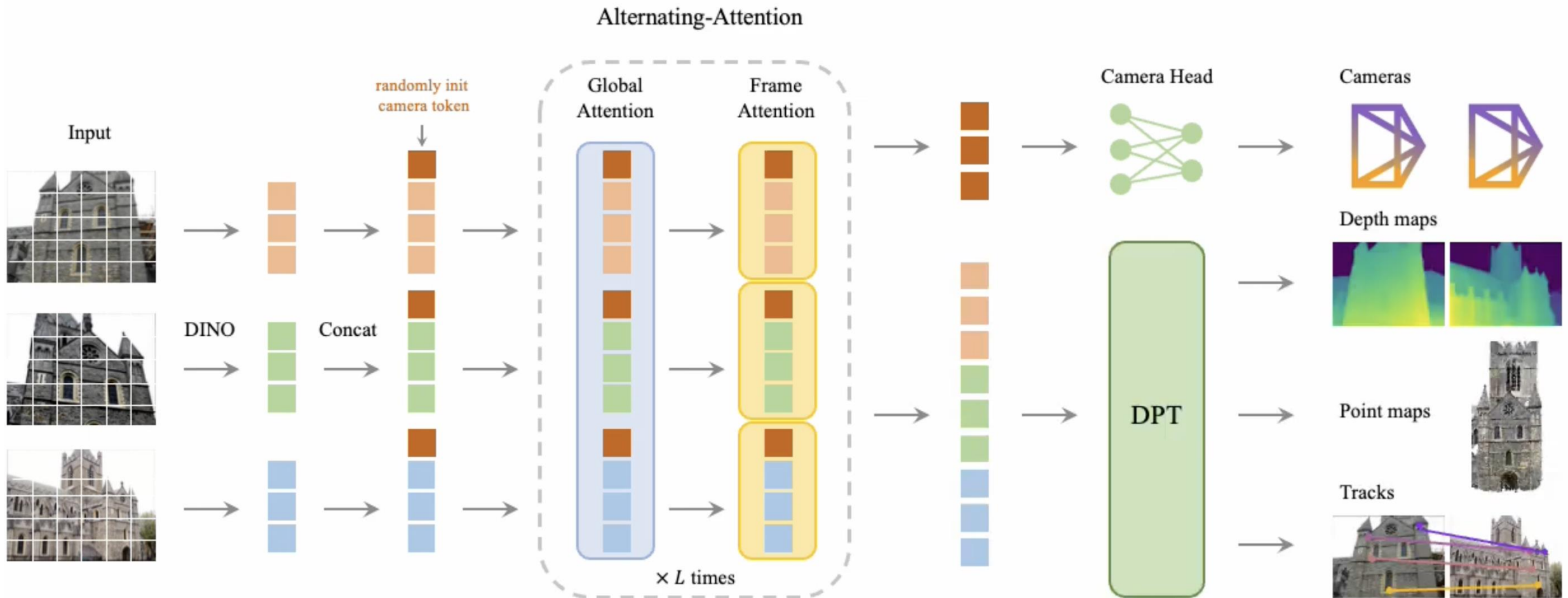


Connecting 2D with 3D



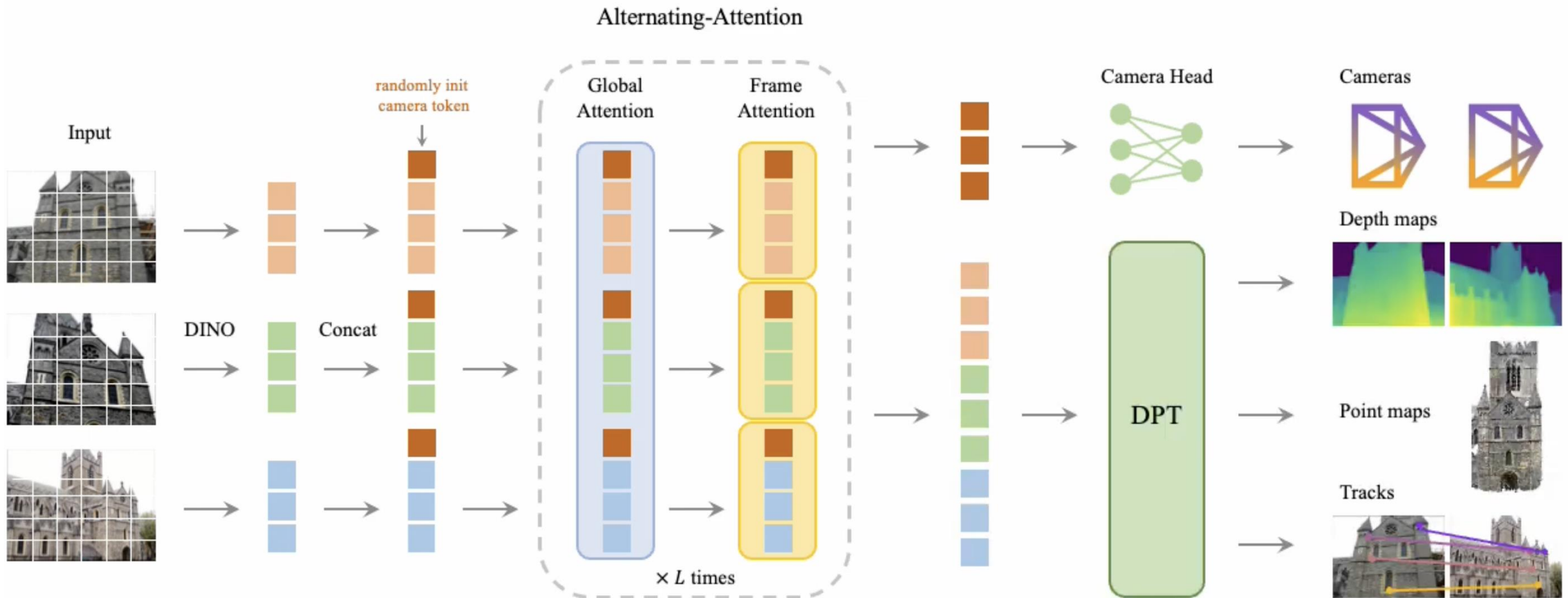
Can we use feed-forward networks to do 3D property prediction?
Most existing methods were optimization-based (e.g., COLMAP)

Connecting 2D with 3D



VGGT with randomly initialized camera tokens can “implicitly” do the optimization in the feed-forwarding process

Connecting 2D with 3D



Combining different outputs (e.g., depths and camera poses) can be more accurate than point map prediction (leveraging camera models)

Today's Agenda

Part 1: Principles for Image Feature Extraction (Cont' from last lecture)

- Relational features and contextual features

Part 2: Principles for 3D Representations

- Surface and volumetric representations

Part 3: Principles for Computations on 3D Representations

- Point processing

Part 4: Applications of 3D Prediction Models

- Navigation and Interaction

Today's Agenda

Part 1: Principles for Image Feature Extraction (Cont' from last lecture)

- Relational features and contextual features

Part 2: Principles for 3D Representations

- Surface and volumetric representations

Part 3: Principles for Computations on 3D Representations

- Point processing

Part 4: Applications of 3D Prediction Models

- Navigation and Interaction

Principles for Image Feature Extraction

Limitation of Convolutional Neural Networks

- By design it only attends to local features

Two Ideas: Relational Features and Global Contexts

Two Ideas: **Relational Features** and Global Contexts

Non-local Neural Networks

Xiaolong Wang^{1,2*}

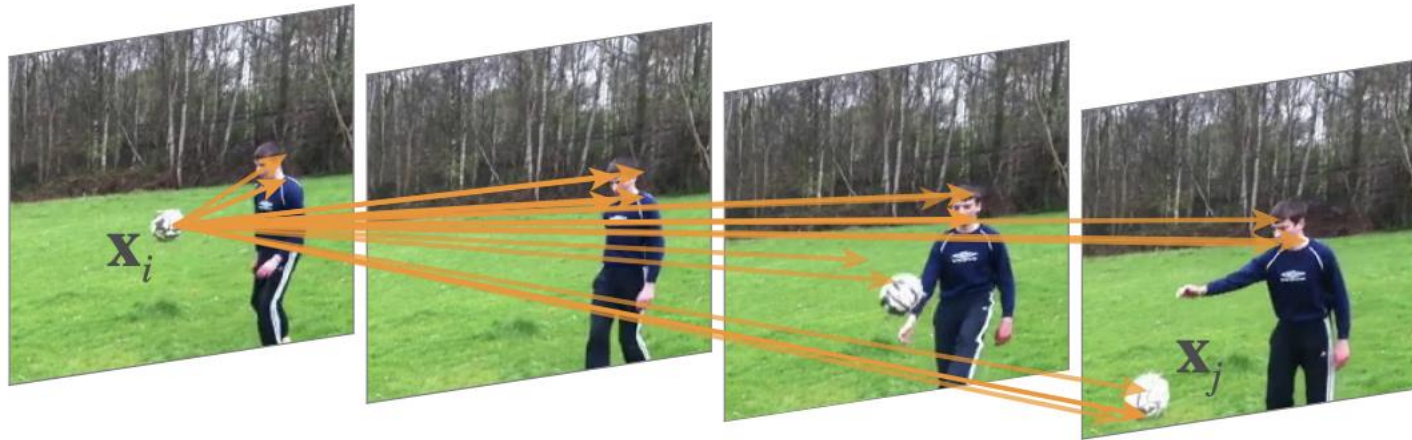
Ross Girshick²

Abhinav Gupta¹

Kaiming He²

¹Carnegie Mellon University

²Facebook AI Research



Motivating Task: Video Activity Classification

- We need to consider the feature of the same object in other frames

Two Ideas: Relational Features and Global Contexts

Non-local Neural Networks

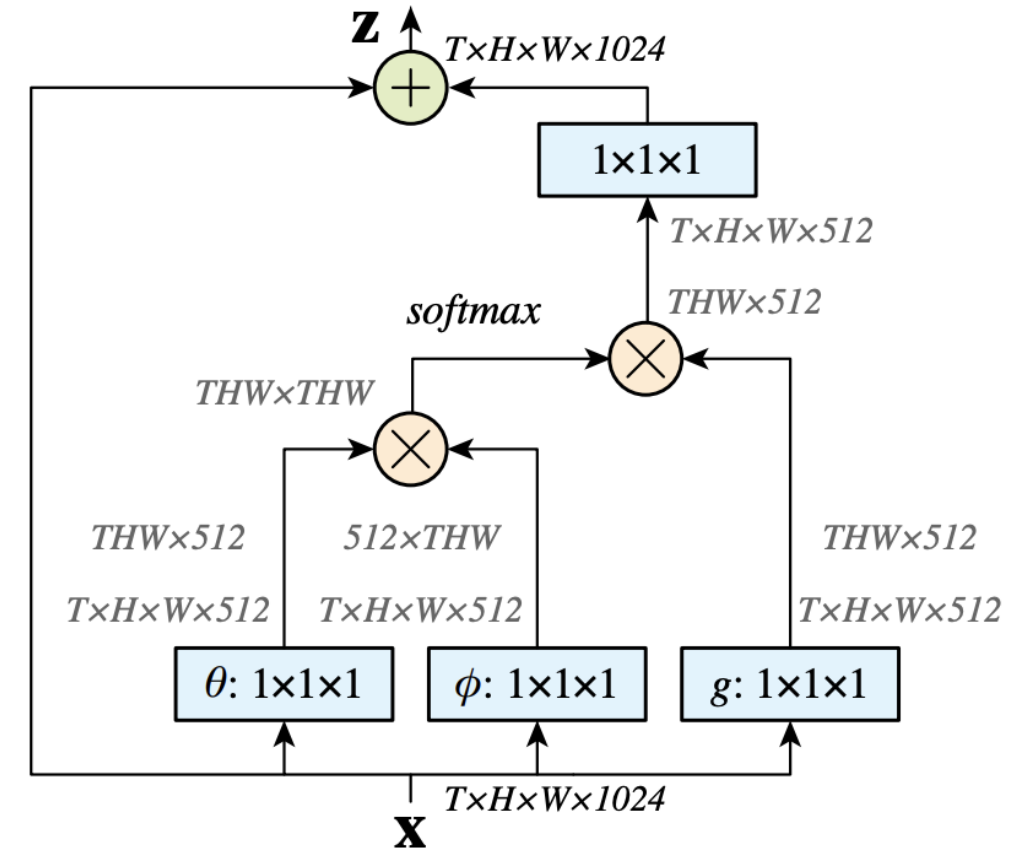
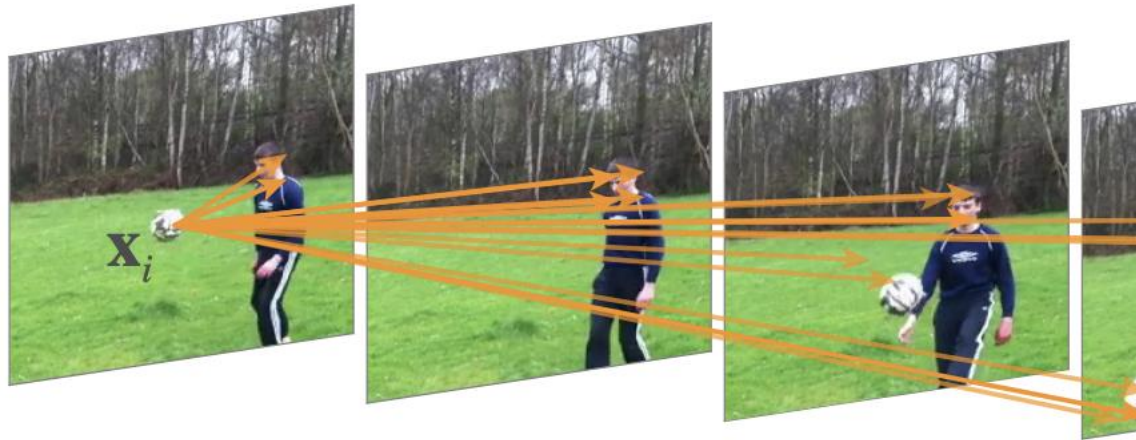
Xiaolong Wang^{1,2*}

Ross Girshick²

Abhinav Gupta¹

¹Carnegie Mellon University

²Facebook AI Research



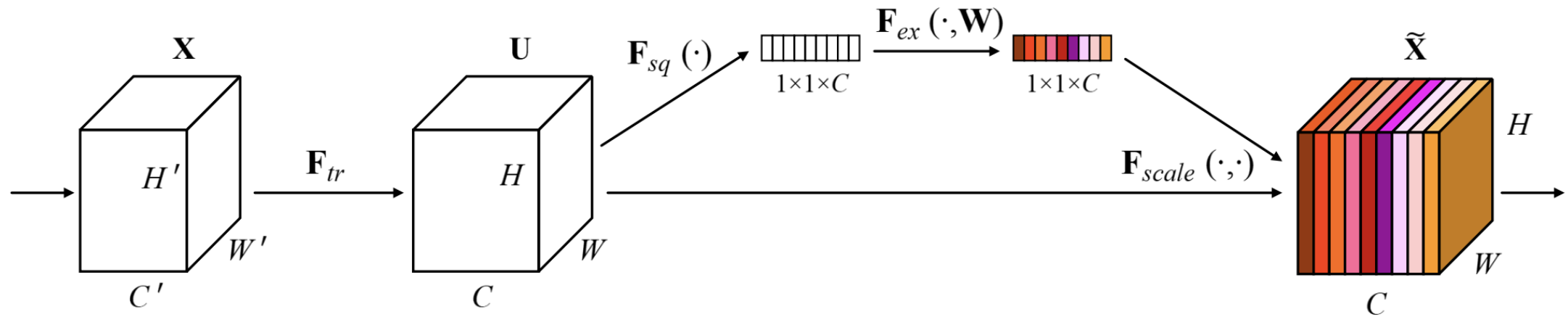
Motivating Task: Video Activity Classification

- We need to consider the feature of the same object in other frames

Two Ideas: Relational Features and Global Contexts

Squeeze-and-Excitation Networks

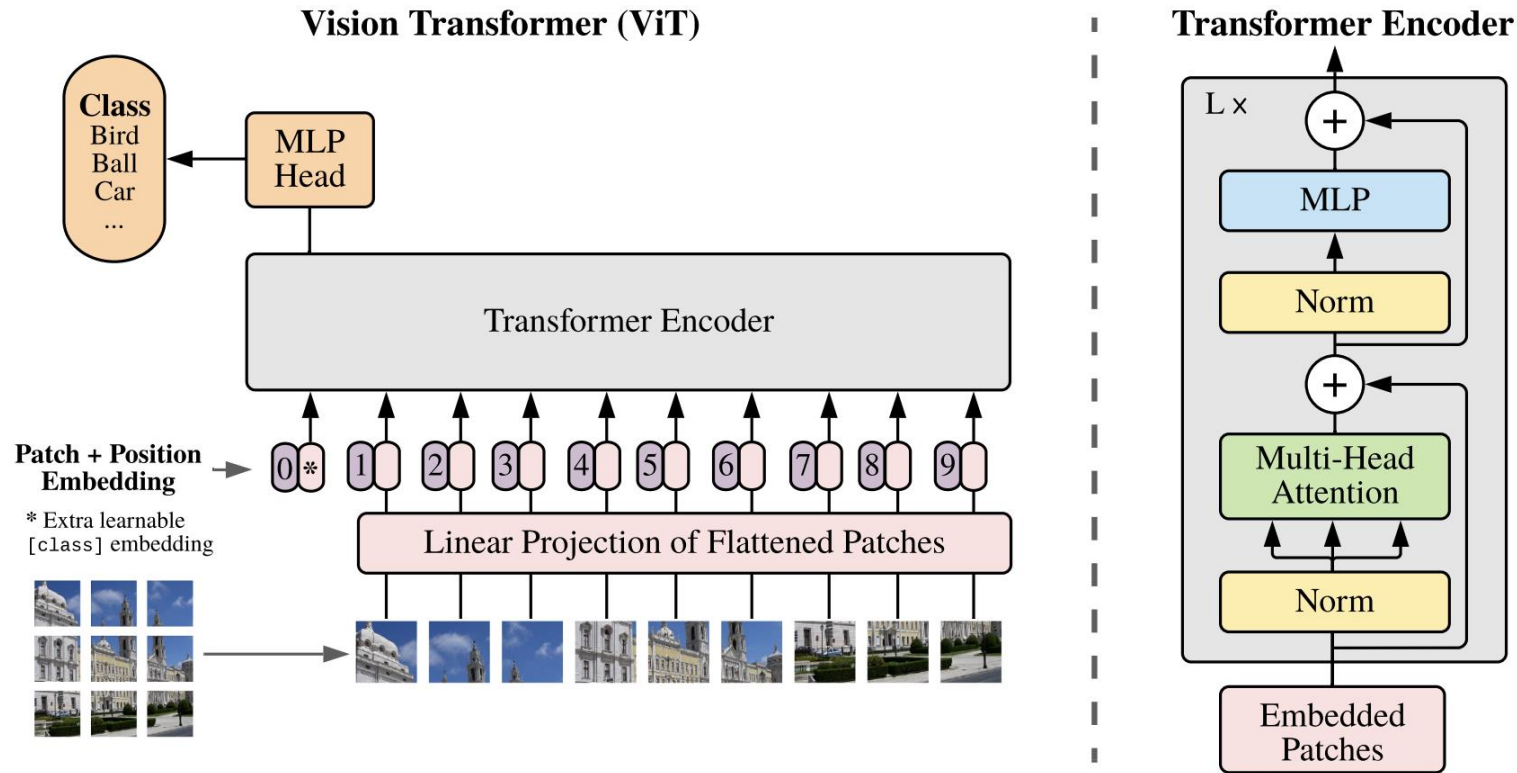
Jie Hu^[0000-0002-5150-1003] Li Shen^[0000-0002-2283-4976] Samuel Albanie^[0000-0001-9736-5134]
Gang Sun^[0000-0001-6913-6799] Enhua Wu^[0000-0002-2174-1428]



Motivating Task: Image Classification and Object Detection

- We need to consider the global image frame for local recognition

Two Ideas: Relational Features and Global Contexts



ViT can be seen as a combination of non-local attentions and global context attention (through the [CLS] token)

Two Ideas: Relational Features and Global Contexts

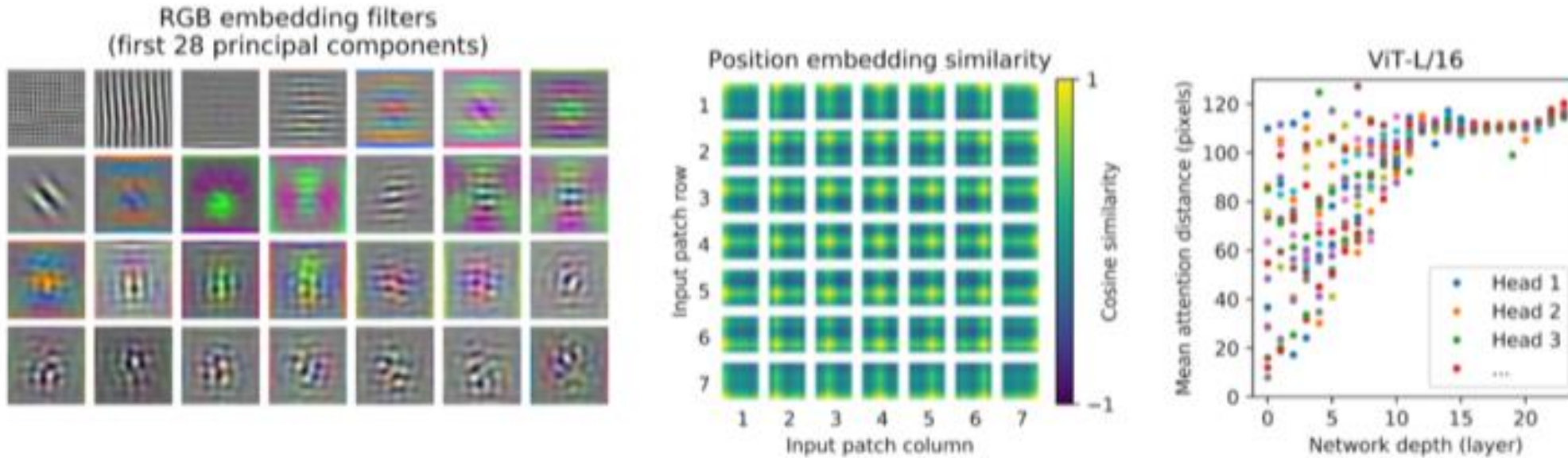
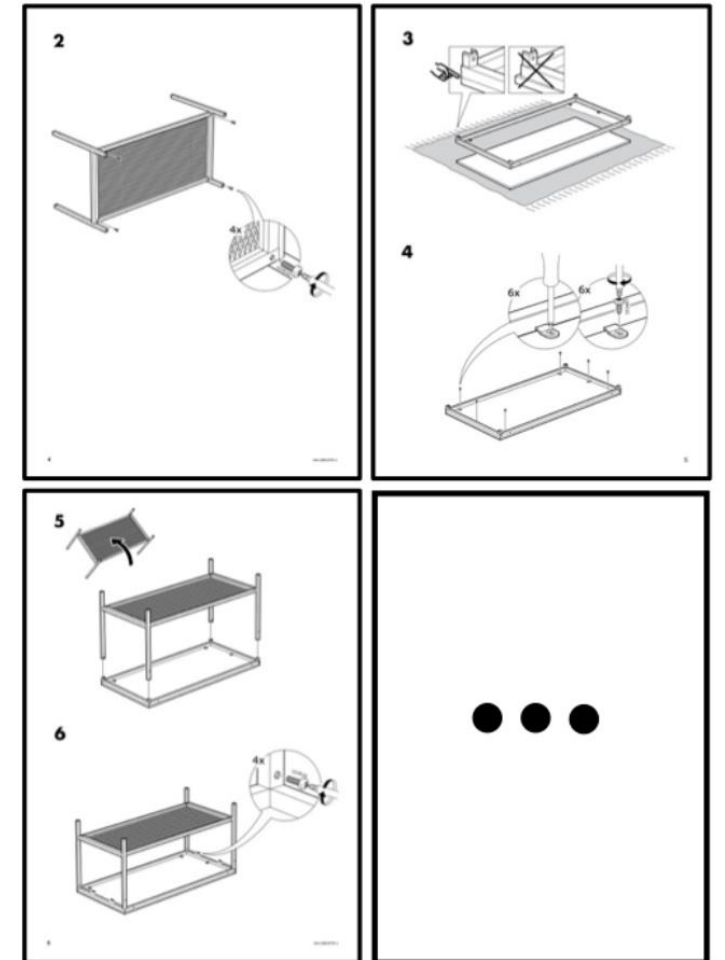


Figure 7: **Left:** Filters of the initial linear embedding of RGB values of ViT-L/32. **Center:** Similarity of position embeddings of ViT-L/32. Tiles show the cosine similarity between the position embedding of the patch with the indicated row and column and the position embeddings of all other patches. **Right:** Size of attended area by head and network depth. Each dot shows the mean attention distance across images for one of 16 heads at one layer. See Appendix [D.7](#) for details.

image taken from "An image is worth 16x16 words"

Notes on Image Representations

- So far we always used perspective projection + RGB images
- There are many other image types
 - Event-based camera
 - Orthographic projection



Today's Agenda

Part 1: Principles for Image Feature Extraction (Cont' from last lecture)

- Relational features and contextual features

Part 2: Principles for 3D Representations

- Surface and volumetric representations

Part 3: Principles for Computations on 3D Representations

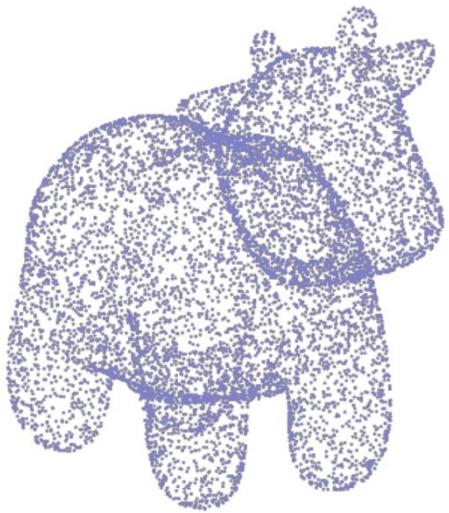
- Point processing

Part 4: Applications of 3D Prediction Models

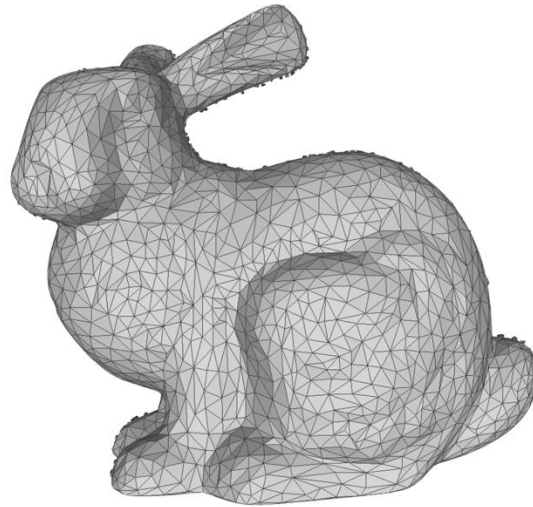
- Navigation and Interaction

3D Representations

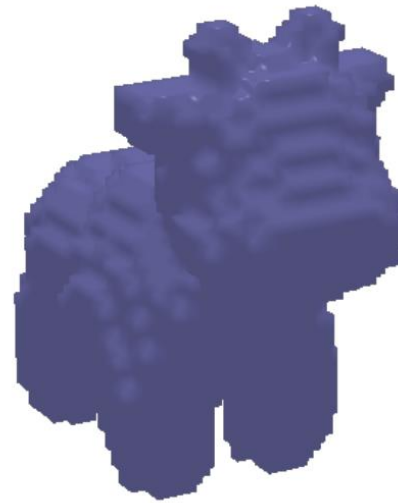
Unlike 2D representations, 3D representations are a lot more diverse



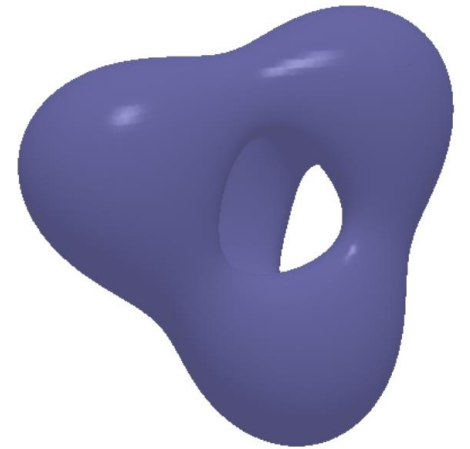
3D Point Cloud



Mesh

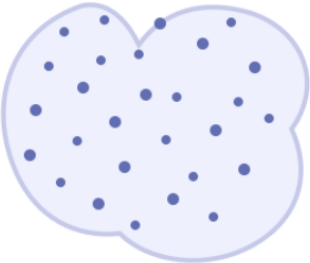


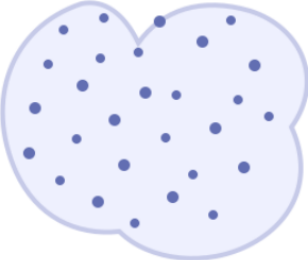
Voxels

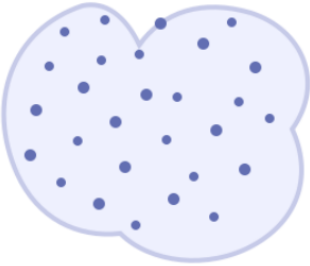


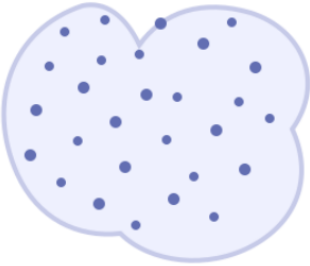
$$\{\mathbf{p} \mid f(\mathbf{p}) = 0\}$$

Implicit Surface

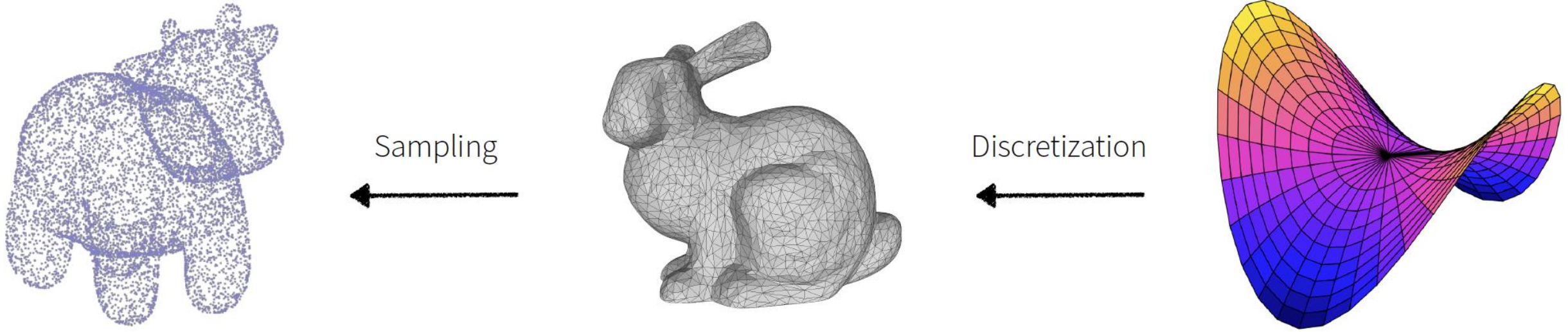
Representation	3D Point Cloud
Visual form	 samples only
Surface vs. volumetric	surface Discrete samples on a surface; no explicit interior or boundary.
Topological structure	 Unordered set; neighborhoods are usually inferred by distance.

Representation	3D Point Cloud	Mesh
Visual form	 samples only	 connected faces
Surface vs. volumetric	surface Discrete samples on a surface; no explicit interior or boundary.	surface Explicit boundary surface made from vertices, edges, and faces.
Topological structure	 Unordered set; neighborhoods are usually inferred by distance.	 Adjacency is explicit; topology comes from the face graph.

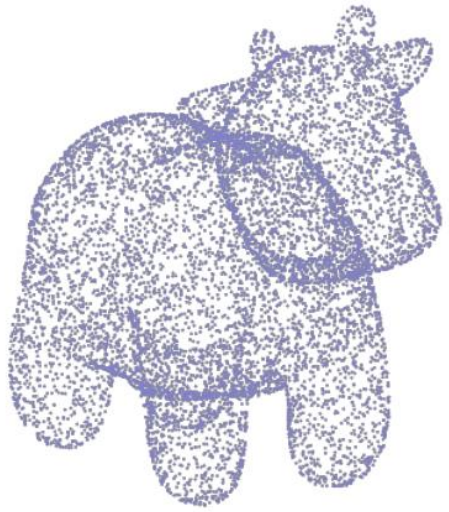
Representation	3D Point Cloud	Mesh	Voxel
Visual form	 samples only	 connected faces	 occupied cells
Surface vs. volumetric	surface Discrete samples on a surface; no explicit interior or boundary.	surface Explicit boundary surface made from vertices, edges, and faces.	volume Samples 3D space itself as occupancy, density, or features.
Topological structure	 Unordered set; neighborhoods are usually inferred by distance.	 Adjacency is explicit; topology comes from the face graph.	 Regular lattice topology; resolution fixes connectivity.

Representation	3D Point Cloud	Mesh	Voxel	Implicit Surface
Visual form	 samples only	 connected faces	 occupied cells	 $\{ p \mid f(p)=0 \}$
Surface vs. volumetric	surface Discrete samples on a surface; no explicit interior or boundary.	surface Explicit boundary surface made from vertices, edges, and faces.	volume Samples 3D space itself as occupancy, density, or features.	continuous A continuous field; the surface is a level set, often with sign.
Topological structure	 Unordered set; neighborhoods are usually inferred by distance.	 Adjacency is explicit; topology comes from the face graph.	 Regular lattice topology; resolution fixes connectivity.	 Topology is implicit and can change with the learned field.

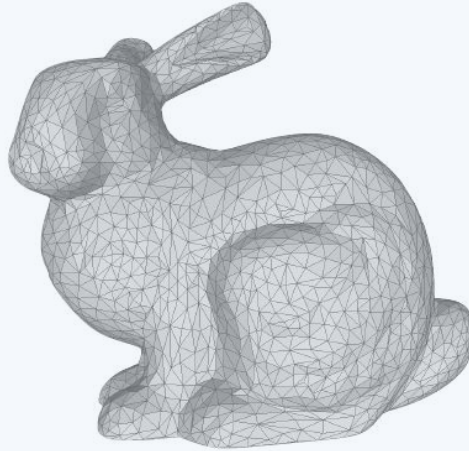
3D Representations: Conversions



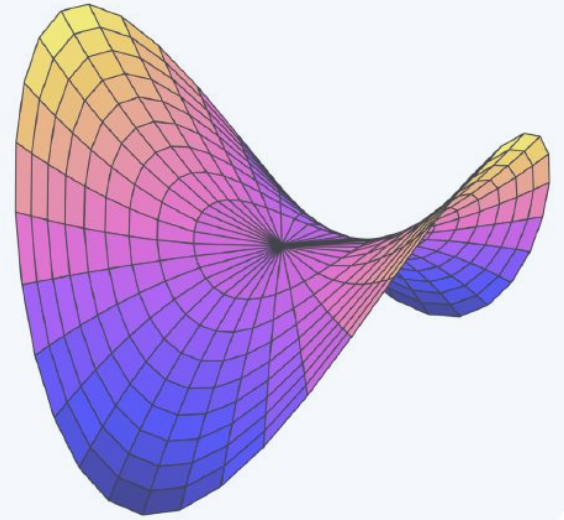
3D Representations: Conversions



Sampling



Discretization



From Implicit Surfaces to Meshes

1. Implicit field



$$S = \{ x \mid f_{\theta}(x) = 0 \}$$

No stored V, F, or E.

- Common strategy: sample points to find the “crossing” points

From Implicit Surfaces to Meshes

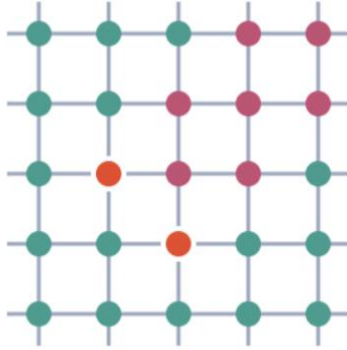
1. Implicit field



$$S = \{ x \mid f_{\theta}(x) = 0 \}$$

No stored V, F, or E.

2. Sample space



Query f on cells.

Opposite signs mean
a surface crossing.

- Common strategy: sample points to find the “crossing” points

From Implicit Surfaces to Meshes

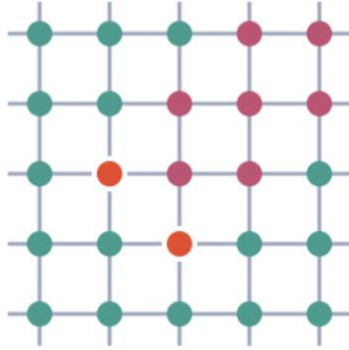
1. Implicit field



$$S = \{ x \mid f_{\theta}(x) = 0 \}$$

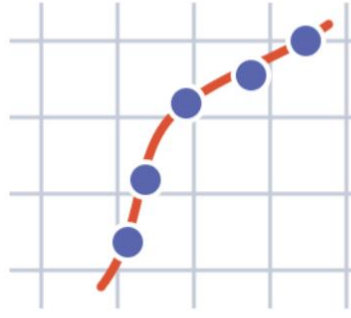
No stored V, F, or E.

2. Sample space



Query f on cells.
Opposite signs mean
a surface crossing.

3. Find vertices



Interpolate each
zero-crossing point.
 $v = x_i + t(x_j - x_i)$

- Common strategy: sample points to find the “crossing” points

From Implicit Surfaces to Meshes

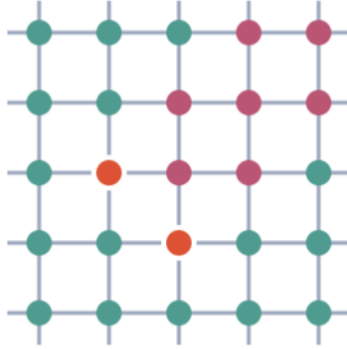
1. Implicit field



$$S = \{ x \mid f_{\theta}(x) = 0 \}$$

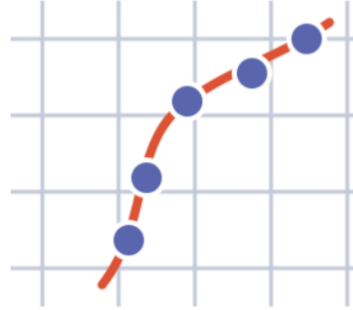
No stored V, F, or E.

2. Sample space



Query f on cells.
Opposite signs mean
a surface crossing.

3. Find vertices



Interpolate each
zero-crossing point.
 $v = x_i + t(x_j - x_i)$

4. Extract faces



Extraction returns
triangles F .
Now connectivity exists.

- Common strategy: sample points to find the “crossing” points

From Implicit Surfaces to Meshes

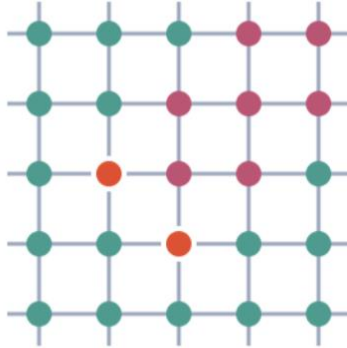
1. Implicit field



$$S = \{ x \mid f_{\theta}(x) = 0 \}$$

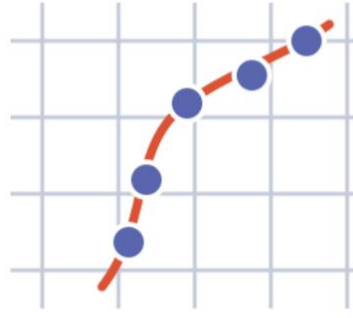
No stored V, F, or E.

2. Sample space



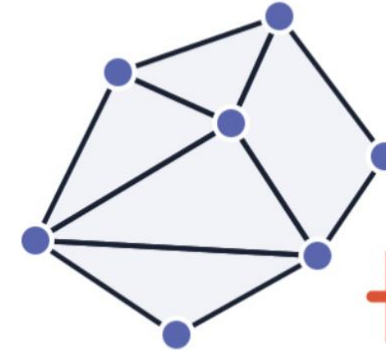
Query f on cells.
Opposite signs mean
a surface crossing.

3. Find vertices



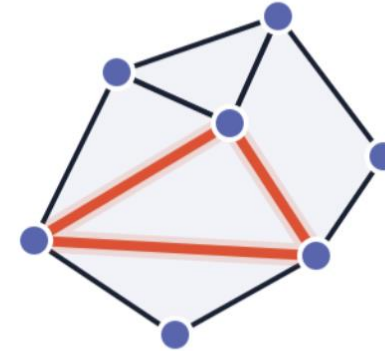
Interpolate each
zero-crossing point.
 $v = x_i + t(x_j - x_i)$

4. Extract faces



Extraction returns
triangles F .
Now connectivity exists.

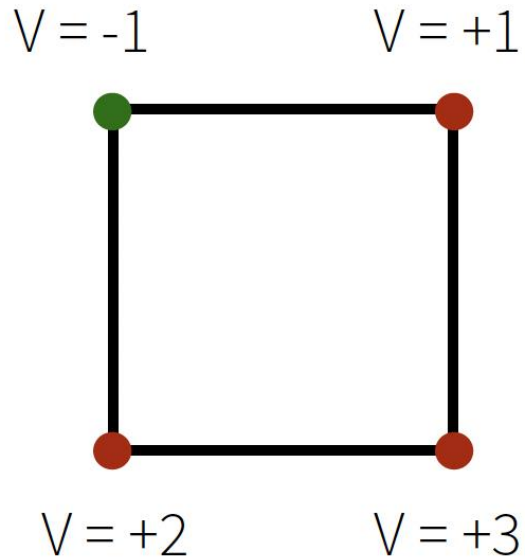
5. Derive edges



Collect every triangle side,
then remove duplicates.

- Common strategy: sample points to find the “crossing” points

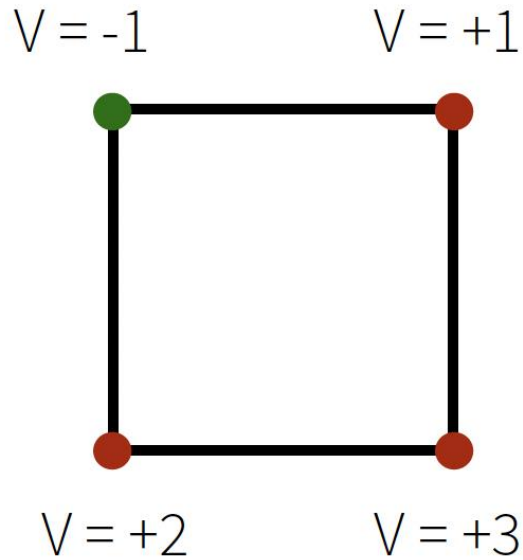
From Signed Vertices to Faces



Where should the surface boundary be?

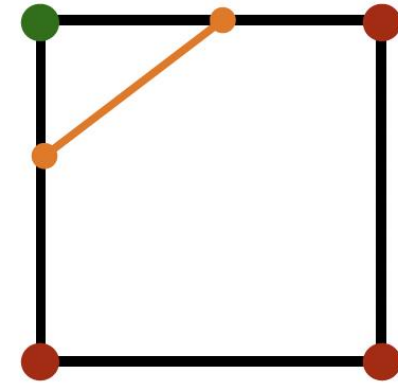
Green — inside, red — outside

From Signed Vertices to Faces



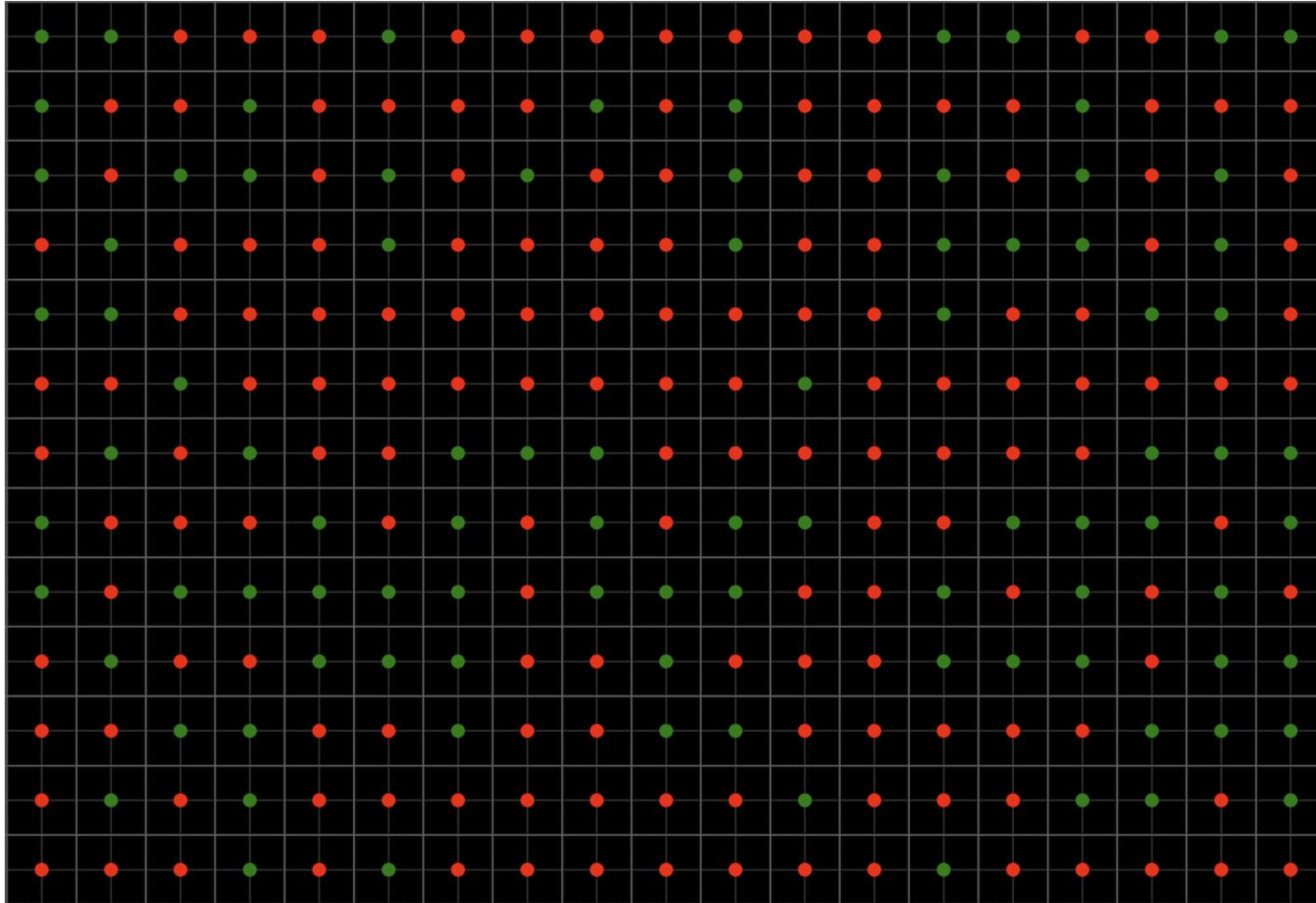
Green — inside, red — outside

Where should the surface boundary be?



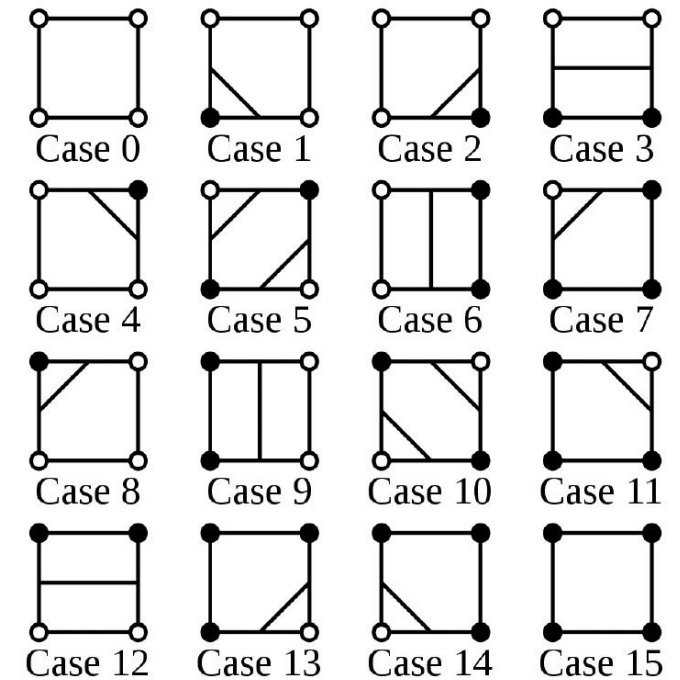
Location of boundary determined by vertex values

From Signed Vertices to Faces: Marching Squares

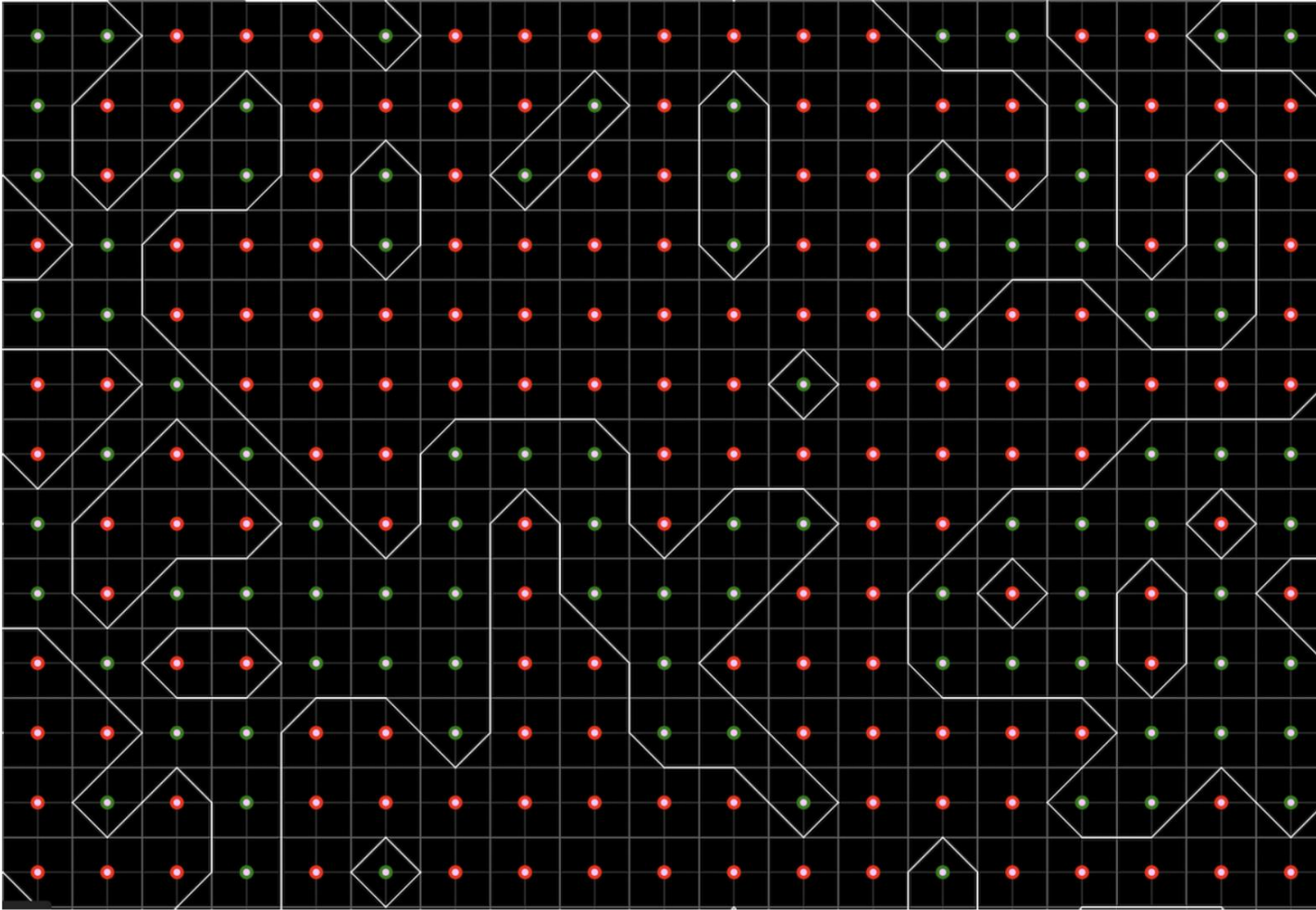


For each cell:

- Use lookup table to draw contours

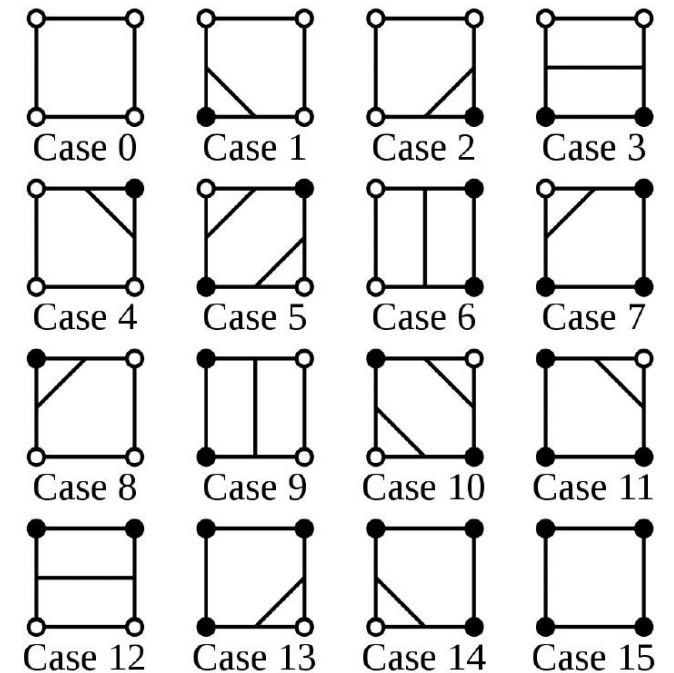


From Signed Vertices to Faces: Marching Squares

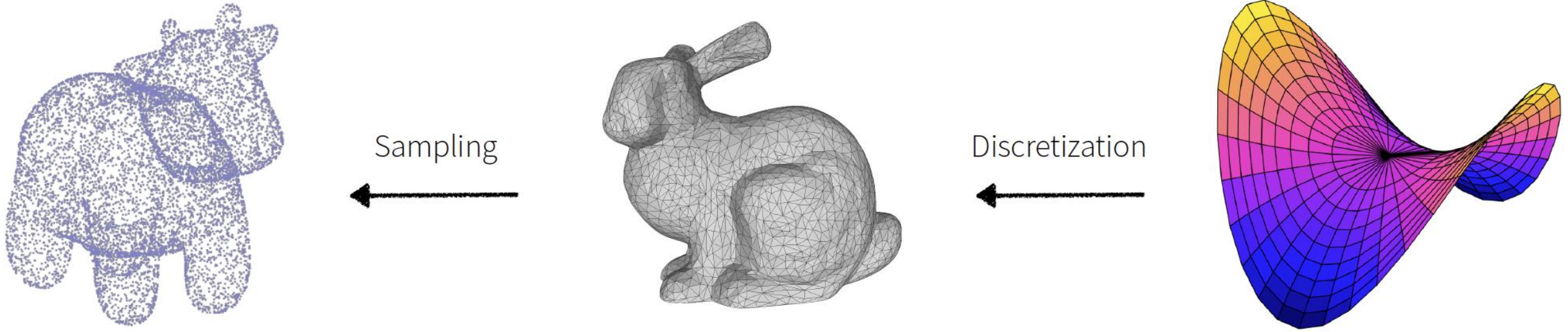


For each cell:

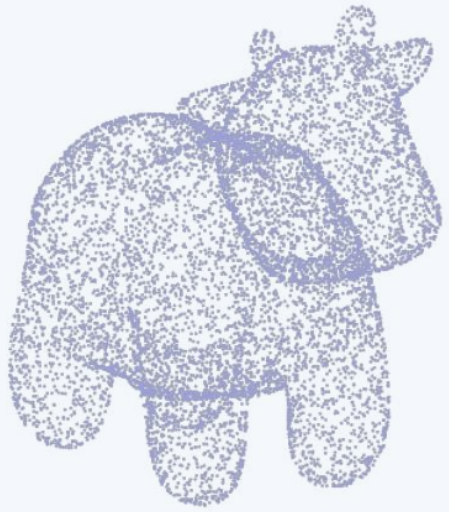
- Use lookup table to draw contours



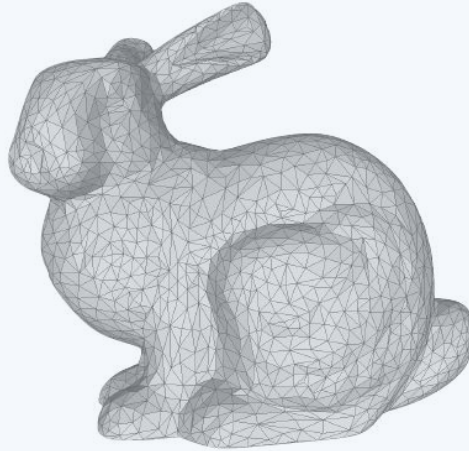
3D Representations: Conversions



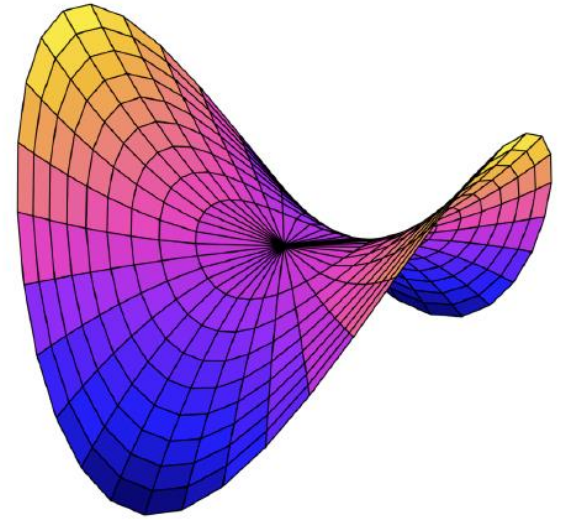
3D Representations: Conversions



Sampling

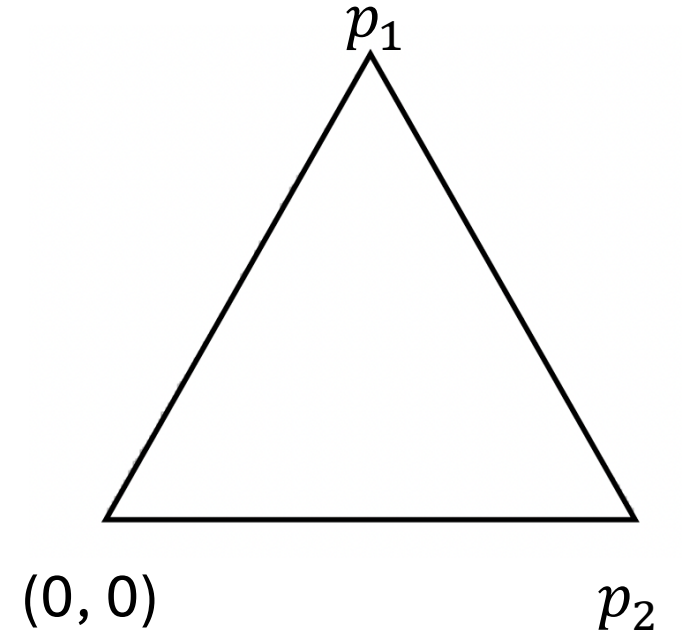


Discretization



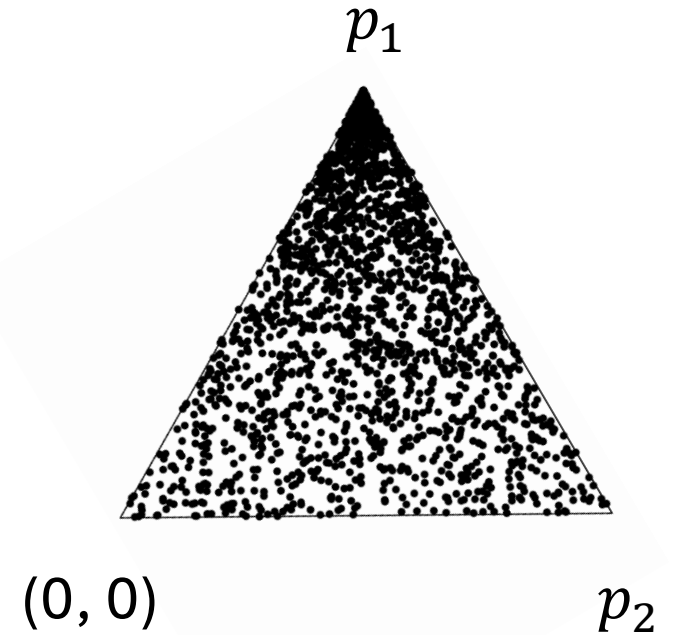
From Meshes to Points

- Step 1: Sample points on triangles
- Key idea: $p = ap_1 + bp_2$
- $a \sim [0, 1]$
- $k \sim [0, 1]$
- $b = k(1 - a)$



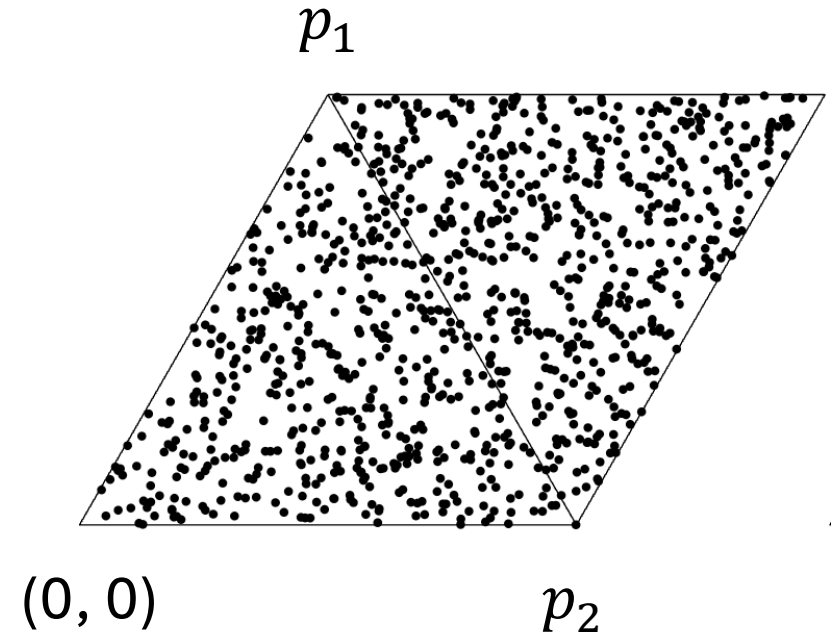
From Meshes to Points

- Step 1: Sample points on triangles
- Key idea: $p = ap_1 + bp_2$
- $a \sim [0, 1]$
- $k \sim [0, 1]$
- $b = k(1 - a)$
- Unfortunately, doesn't work



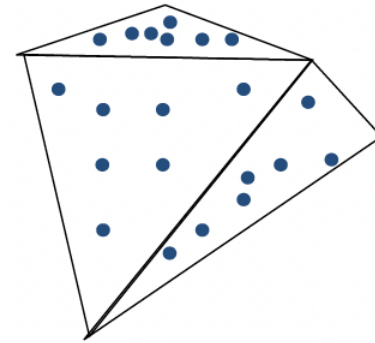
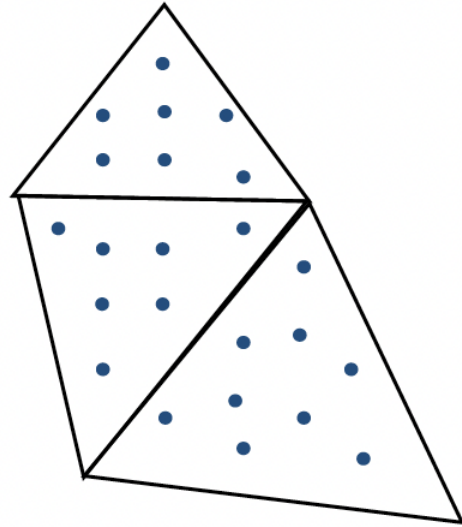
From Meshes to Points

- Step 1: Sample points on triangles
- Key idea: $p = ap_1 + bp_2$
- $a \sim [0, 1]$
- $b \sim [0, 1]$
- This gives you a quadrilateral
- Then we can discard points outside the triangle



From Meshes to Points

- Step 1: Sample triangles



- Which one is better?
- Usually we do “uniform” by area

Notes and Take-Aways

- “Correctly” converting different representations is challenging
 - Many hidden details: discretization resolution, thresholds, sampling strategies
- Voxel in theory contains “all” information but is very expensive
 - Also hard to obtain from camera signals
- Marching cubes, discretization, and sampling are basic ideas
- Implicit surfaces are good for rendering but not very easy to do computations on

Today's Agenda

Part 1: Principles for Image Feature Extraction (Cont' from last lecture)

- Relational features and contextual features

Part 2: Principles for 3D Representations

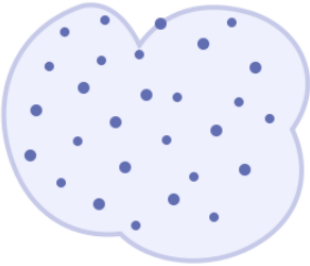
- Surface and volumetric representations

Part 3: Principles for Computations on 3D Representations

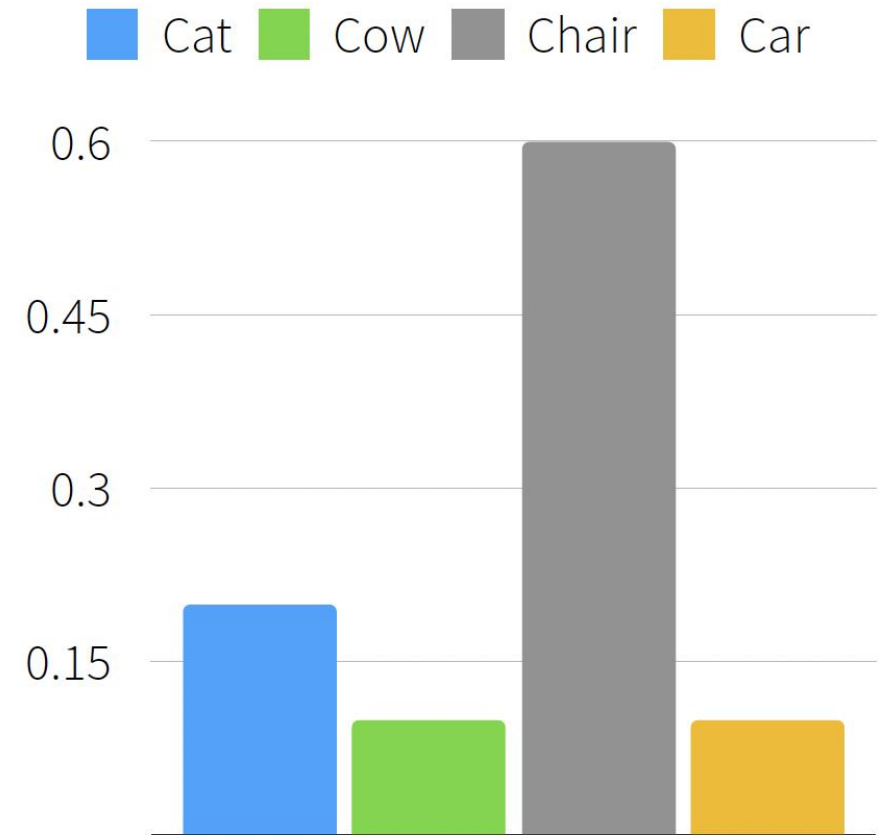
- Point processing

Part 4: Applications of 3D Prediction Models

- Navigation and Interaction

Representation	3D Point Cloud	Mesh	Voxel	Implicit Surface
Visual form	 samples only	 connected faces	 occupied cells	 $\{ p \mid f(p)=0 \}$
Surface vs. volumetric	surface Discrete samples on a surface; no explicit interior or boundary.	surface Explicit boundary surface made from vertices, edges, and faces.	volume Samples 3D space itself as occupancy, density, or features.	continuous A continuous field; the surface is a level set, often with sign.
Topological structure	 Unordered set; neighborhoods are usually inferred by distance.	 Adjacency is explicit; topology comes from the face graph.	 Regular lattice topology; resolution fixes connectivity.	 Topology is implicit and can change with the learned field.
Computation model	Point-wise networks, pooling, or kNN/graph neighborhoods.	Operators on vertices/faces; message passing or geometry ops.	3D convolutions, sparse convs,	

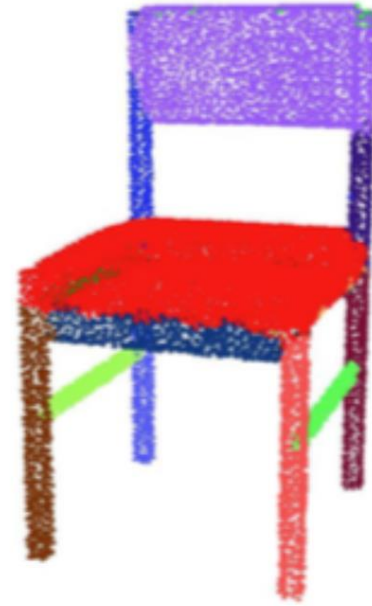
Computation on Point Clouds: Classification



Computation on Point Clouds: Segmentation



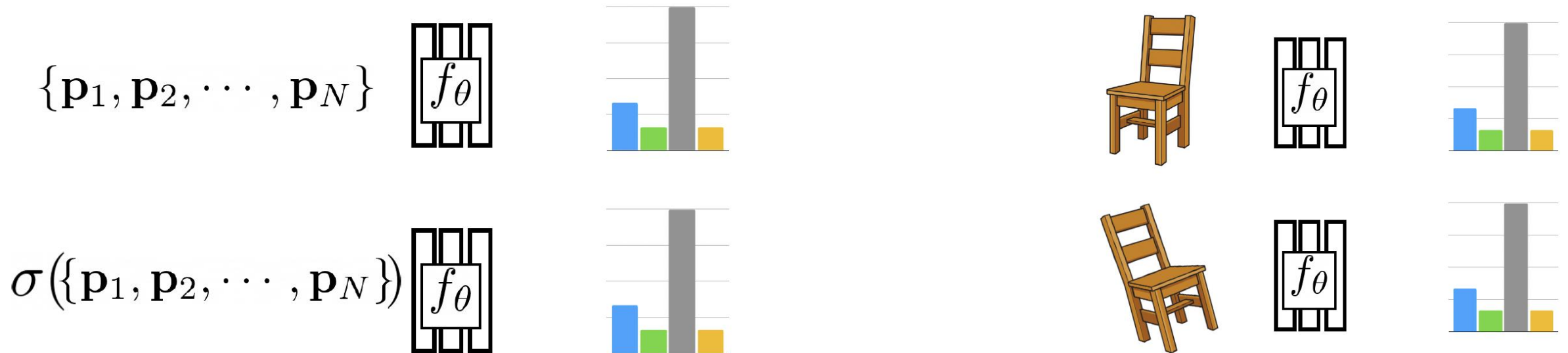
Input: $N \times 3$



Output: $N \times C$

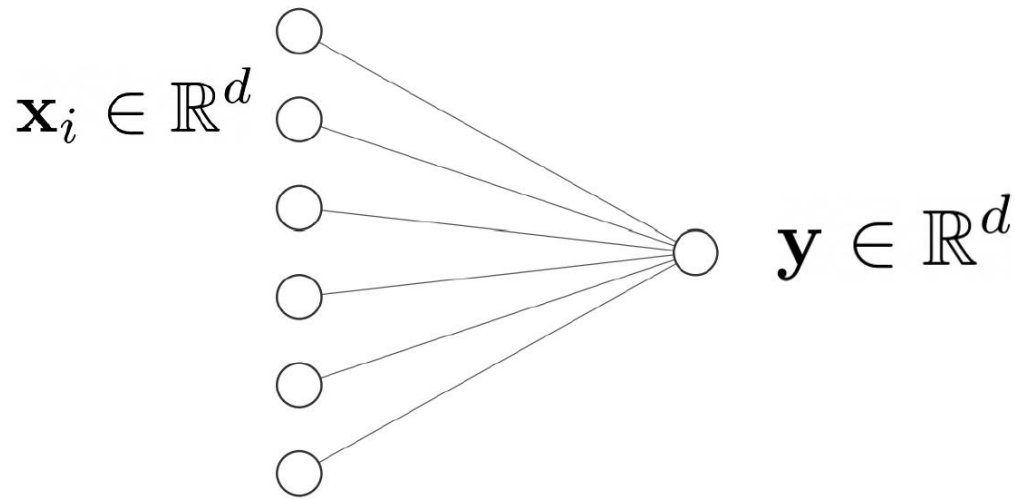
The Design Principles of Point Cloud Networks

- Point clouds are not ordered (so convolution does not make sense)
- Therefore, our computational models should be permutation-invariant
- There are other properties we may want (e.g., rotation-invariance)



Two Basic Permutation-Invariant Operations

- Pooling (mean, max) operations are permutation-invariant

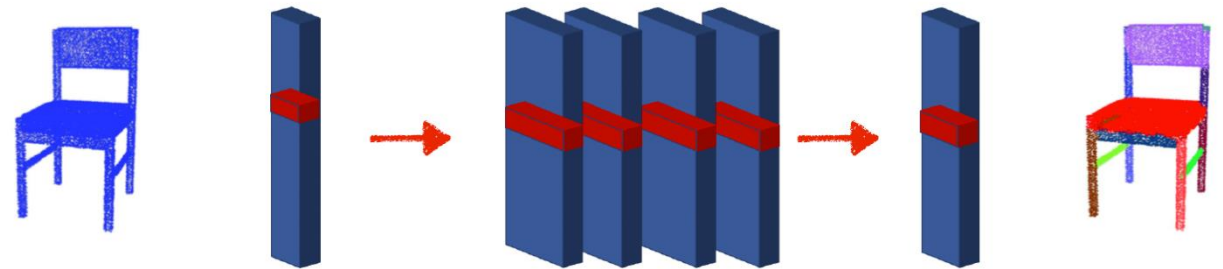
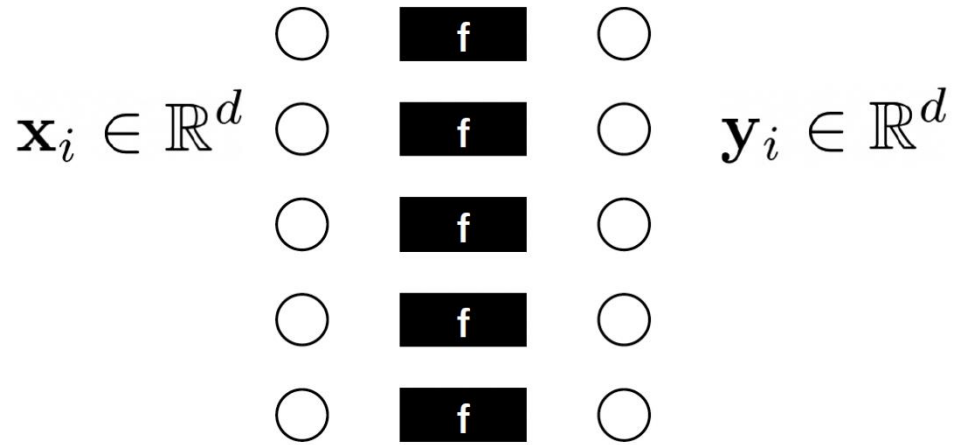


$$y = \text{mean}(\{\mathbf{x}_i\})$$

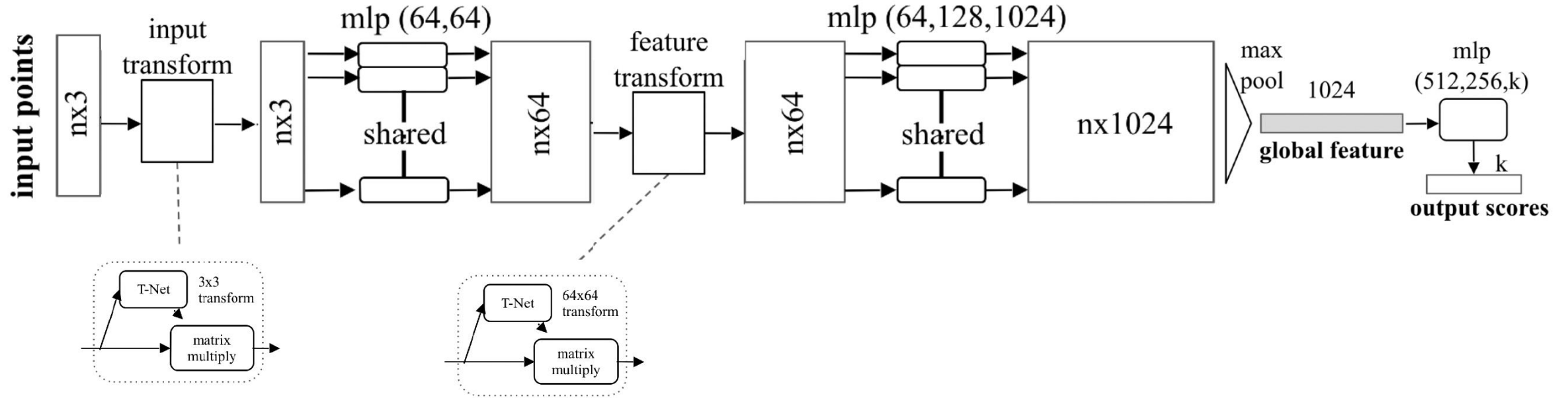
$$y = [\cdots, \max(\{\mathbf{x}_i^{\mathbf{d}}\}), \cdots]$$

Two Basic Permutation-Invariant Operations

- Pooling (mean, max) operations are permutation-invariant
- Applying the same function to each point is permutation-invariant

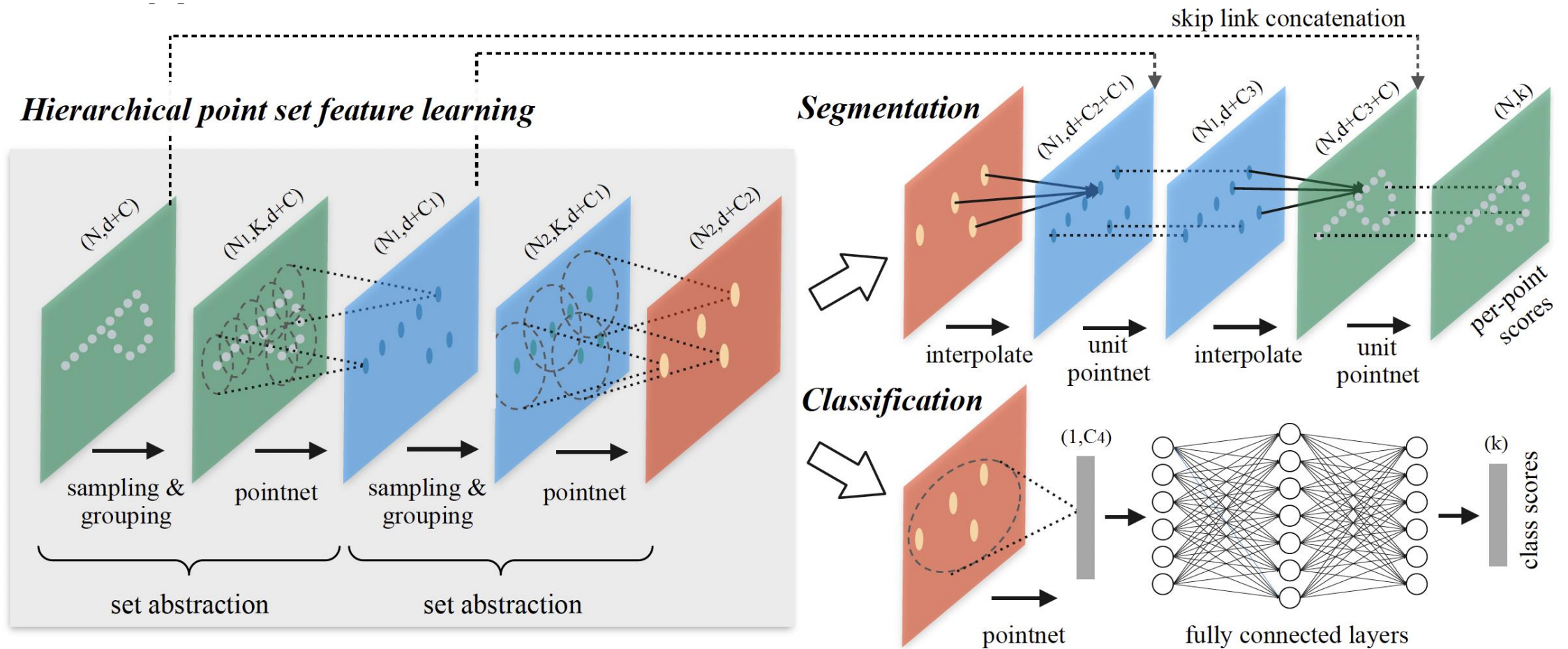


Classification with PointNet



- “Transform” layers is essentially a squeeze-excitation network!
- A geometric motivation: “undo” the rotation

Classification with PointNet++



- Key idea: adding locality structures through distance-based grouping

Principles of Point Cloud Computation

- Permutation invariance
 - Adding global contexts
 - Adding locality structures
-
- There are other things to consider: rotation-invariance, surface normal structures
 - Also a lot of parameters to tune: resolution, sampling strategies

Today's Agenda

Part 1: Principles for Image Feature Extraction (Cont' from last lecture)

- Relational features and contextual features

Part 2: Principles for 3D Representations

- Surface and volumetric representations

Part 3: Principles for Computations on 3D Representations

- Point processing

Part 4: Applications of 3D Prediction Models

- Navigation and Interaction

Applying 3D Models for Robotics Applications

- Two challenges: large scenes and action-relevant representations

Applying 3D Models to Large Scenes

- Running PointNet on Millions of Points is too heavy
- Use hierarchical models
- These models are usually heavily biased by the application (social navigation and object rearrangements)
- But in general: doing “attention” on a local structure is a powerful idea

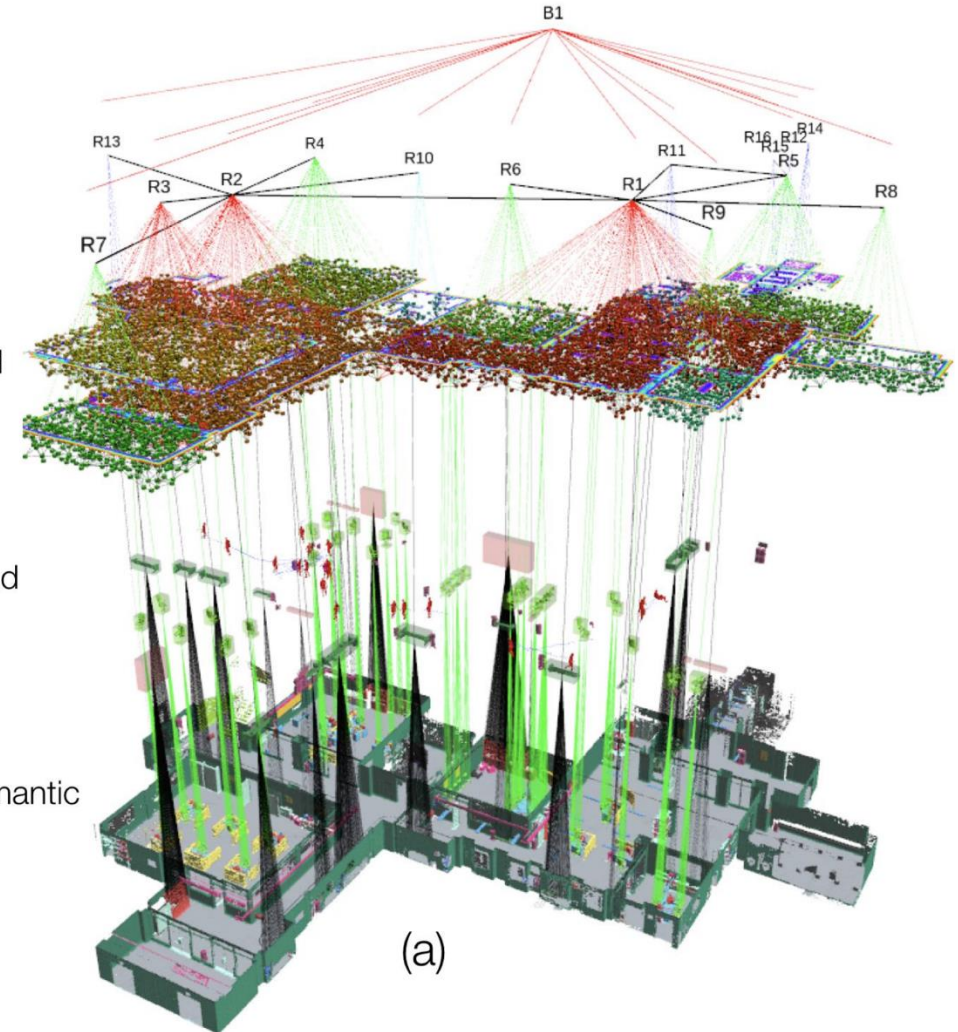
Layer 5:
Buildings

Layer 4:
Rooms

Layer 3:
Places and
Structures

Layer 2:
Objects and
Agents

Layer 1:
Metric-Semantic
Mesh



Applying 3D Models to Large Scenes

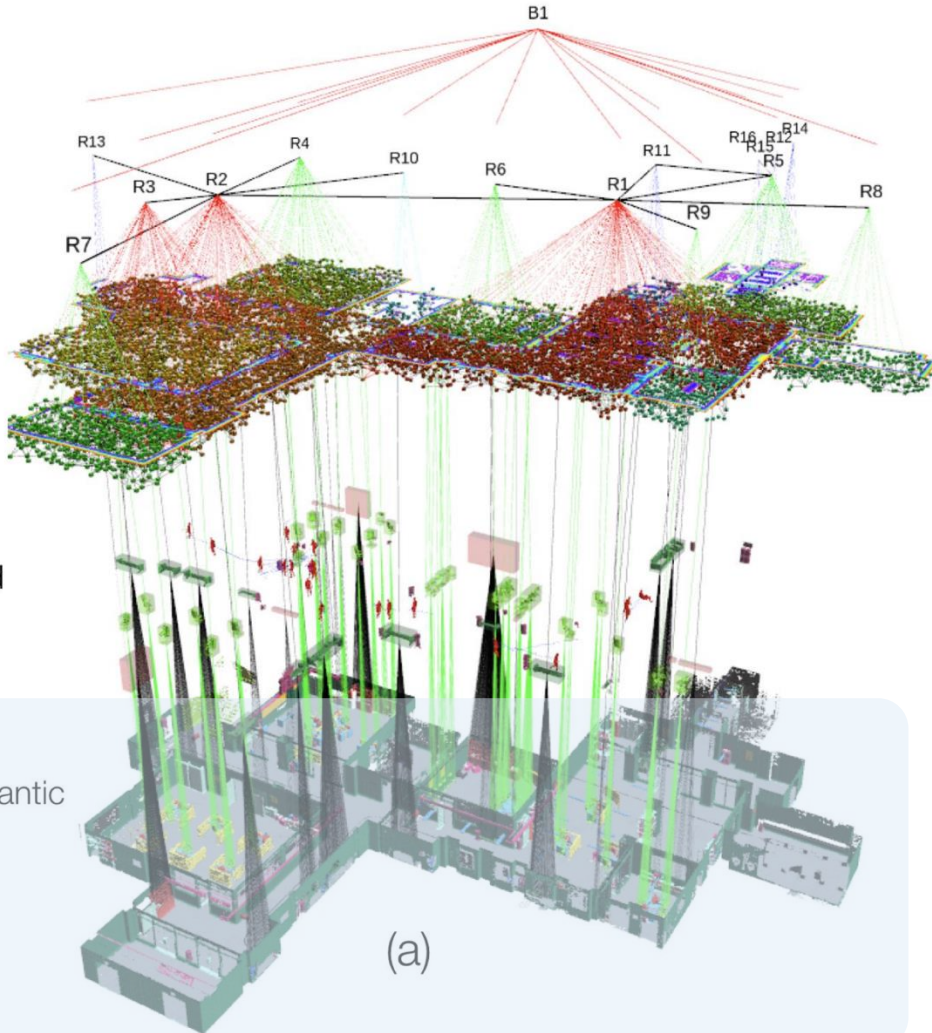
Layer 5:
Buildings

Layer 4:
Rooms

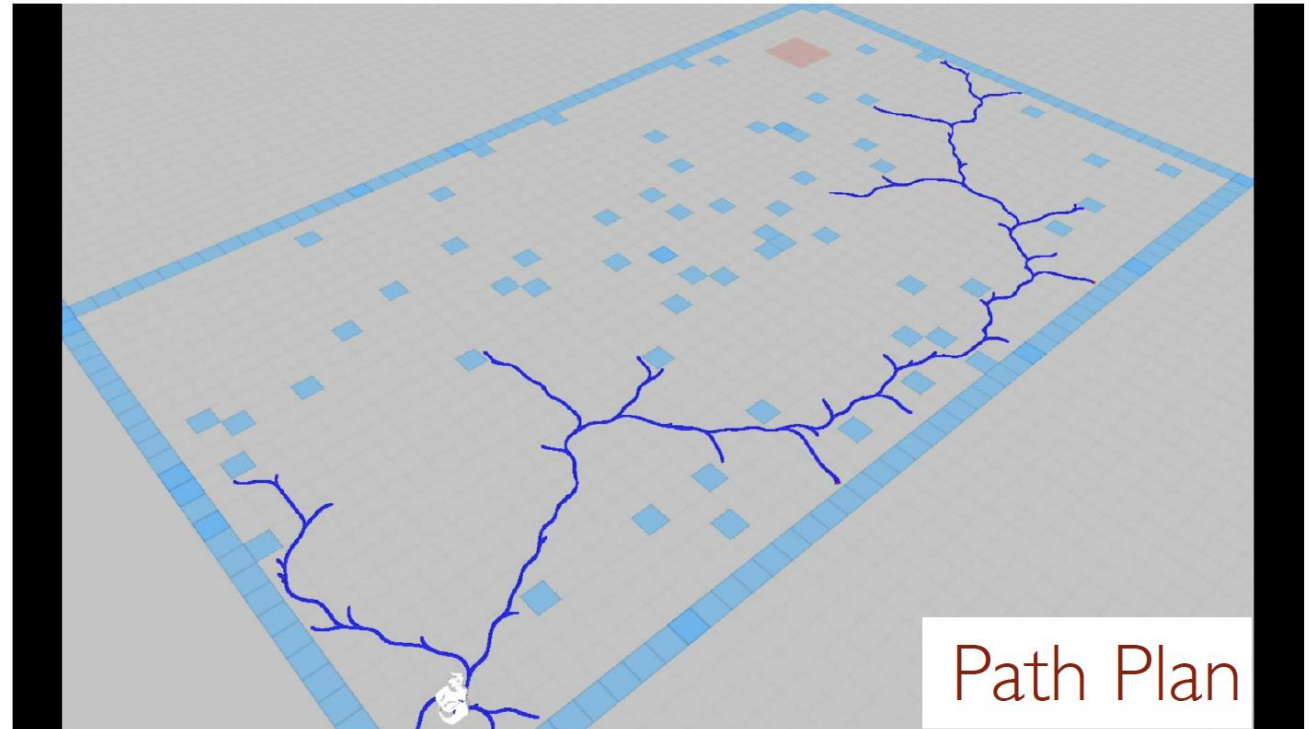
Layer 3:
Places and
Structures

Layer 2:
Objects and
Agents

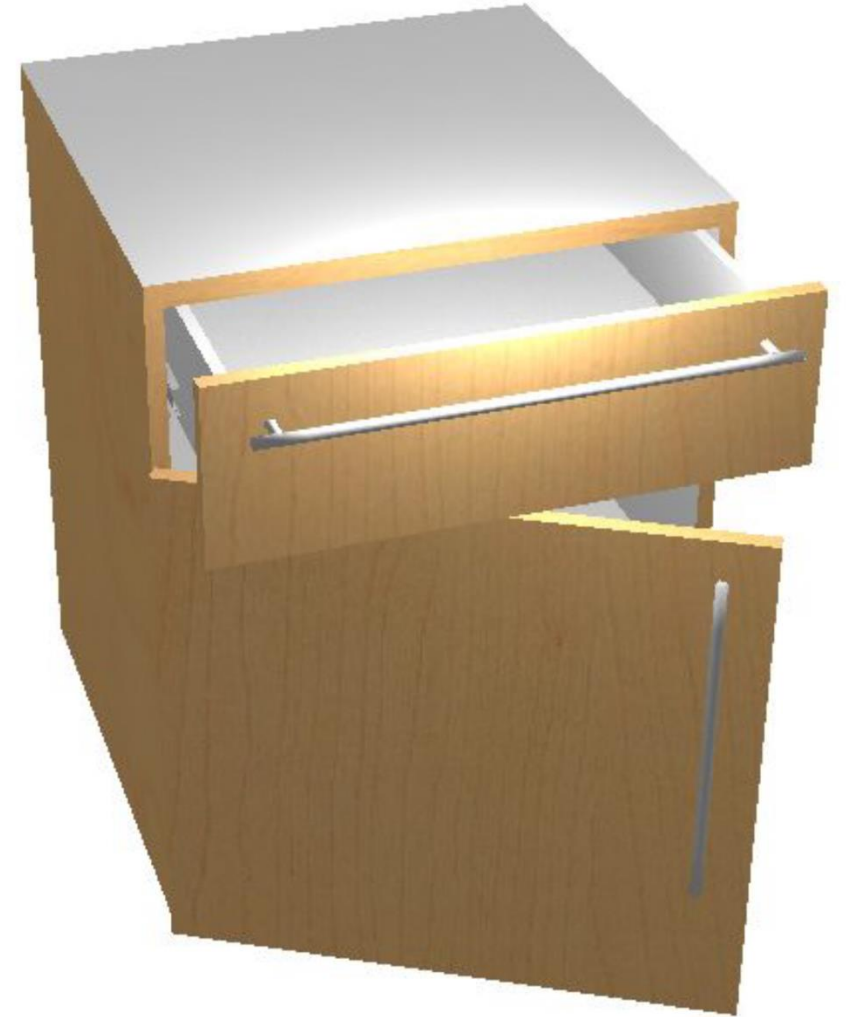
Layer 1:
Metric-Semantic
Mesh



- Mesh makes collision checking easy and subsequently, path planning easy

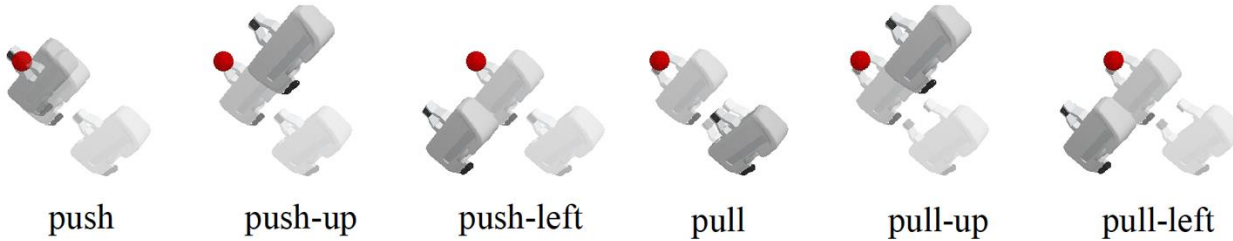


Applying 3D Models for Affordance Prediction

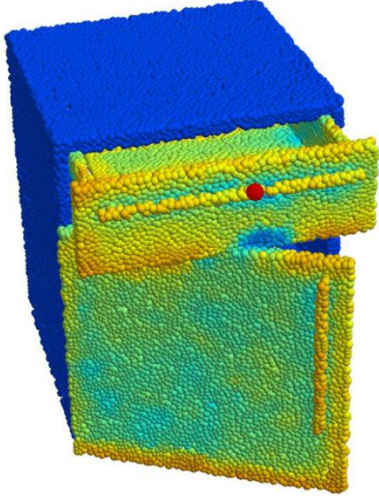


Applying 3D Models for Affordance Prediction

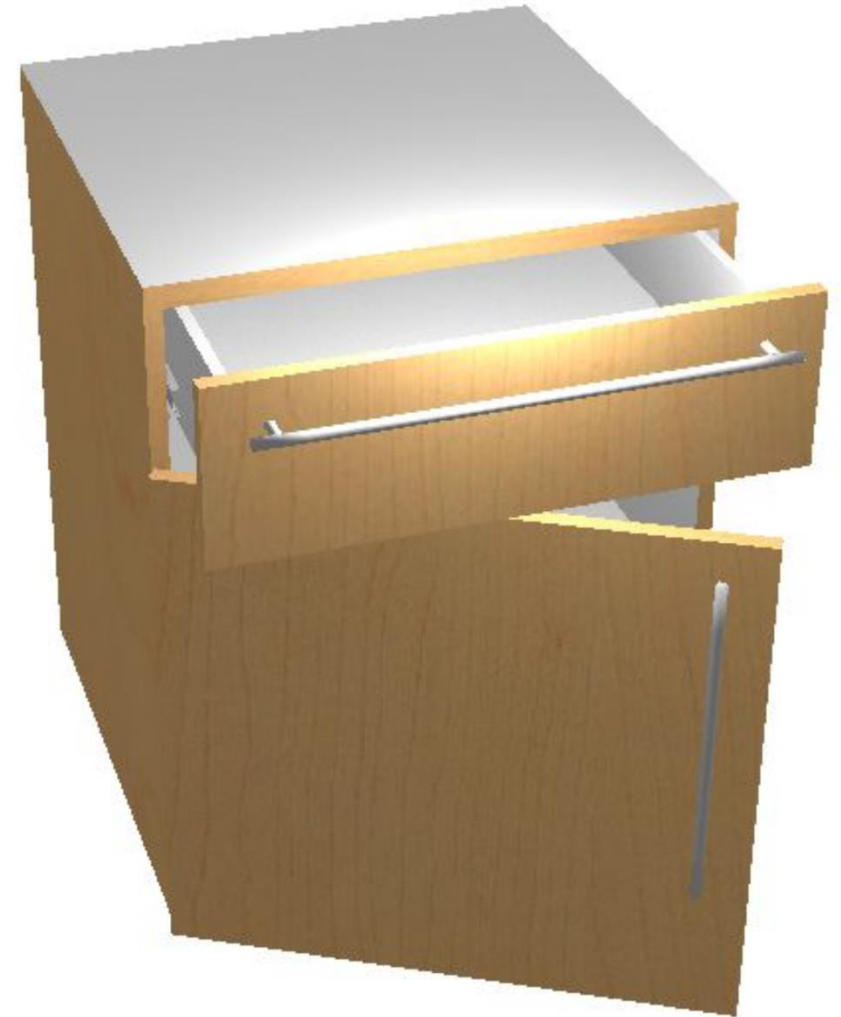
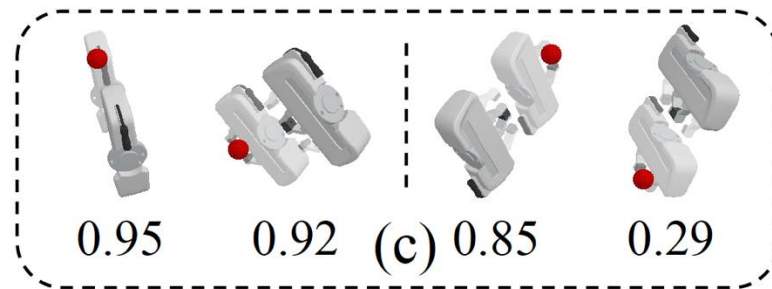
What?



Where?



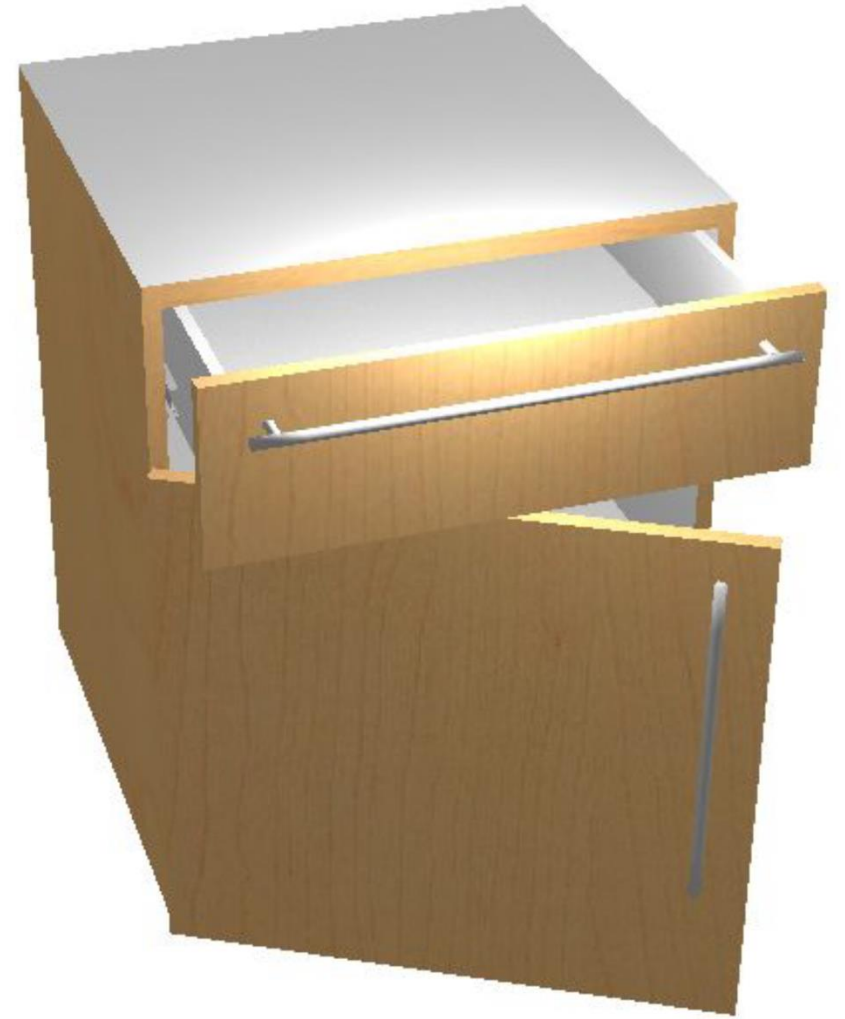
How?



What actions can a robot take?

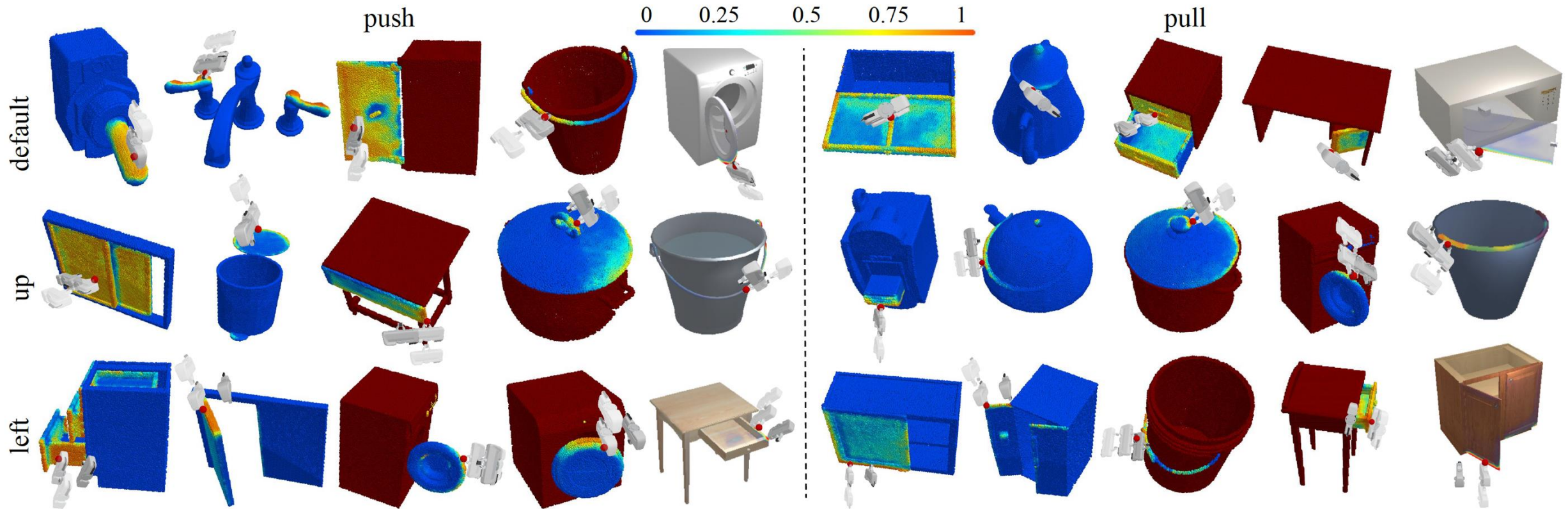
Applying 3D Models for Affordance Prediction

- Use PointNet to predict per-point action candidates



Applying 3D Models for Affordance Prediction

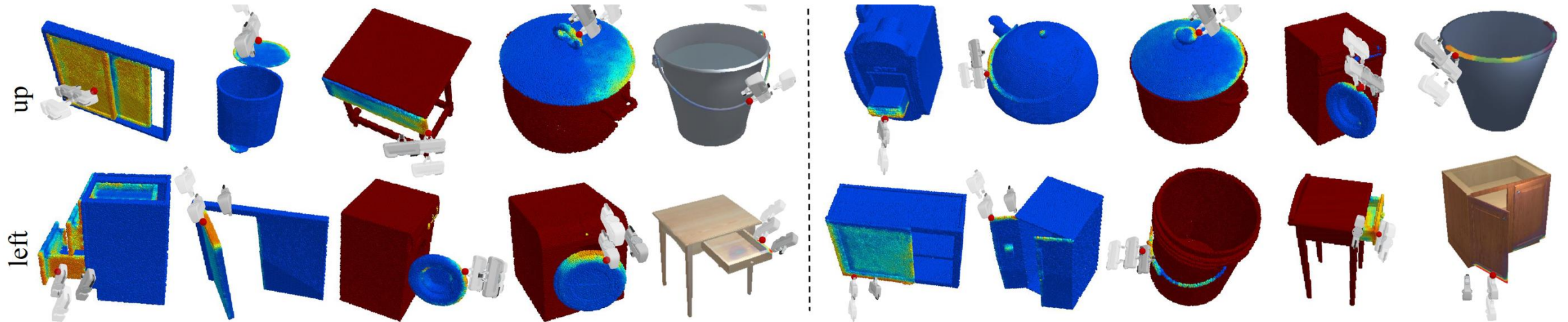
- Use PointNet to predict per-point action candidates



Blue: seen categories, Red: Novel categories

Applying 3D Models for Affordance Prediction

- Use PointNet to predict per-point action candidates
- For now these are just “proposals”
- We need an action selection module to decide where and how to act



Blue: seen categories, Red: Novel categories

Questions for Discussion

- Given that we already have depth sensors and IMU sensors, what are the advantages VGGT-like 2D-to-3D reconstruction methods?
- For human-computer interaction purposes, what should be design principles of 3D representations and computational methods?
 - visualization, editing, simulation
- For your application, what is the “best” 3D representation? What are important hyperparameters (e.g., resolutions)?
- What are limitations of PointNet / PointNet++ as a 3D computational model?